

Appendix 7. Input parameters and their definition

Abbreviations used in parameter names

No	Acronym	Definition	No	Acronym	Definition
1.	AF	"Agroforestry Zone" – overall design of the system	14.	Mn	Nutrients in Litter Layer
2.	C	Crop (C = Crop, C_N = Crop Nutrient or CW = Crop Water)	15.	Mn2	Nutrients in Soil Organic Matter (SOM)
3.	Ca	Crop Calendar (schedule)	16.	N	Nutrient (currently including N and P)
4.	Cent	Input Output Summary for Litter (based on Century Model)	17.	P	Profitability (economic sector of the model)
5.	Cent2	Input Output Summary for SOM (based on Century Model)	18.	PD	Pest and Disease
6.	Cq	Crop Sequence (crop parameters)	19.	Rain	Rain
7.	E	Erosion	20.	Rt	Root
8.	Evap	Evaporation	21.	S&B	Slash and Burn
9.	G	Grazing	22.	S	Soil Structure
10.	LF	Lateral Flow	23.	T	Tree (T=Tree, T_N=Tree Nutrient or TW=Tree Water)
11.	Light	Light	24.	TF	Tree Fruit (oil Palm Module)
12.	Mc	Carbon in Litter Layer	25.	Temp	Temperature
13.	Mc2	Carbon in Soil Organic Matter (SOM)	26.	W	Water

Note: Without green dot are CORE modules, with '•' (green dot) in front means ADDITIONAL modules

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
1.	AF_AnyTrees?	Parameter governing an option to simulate system with trees. Value 0 means system without trees, value 1 means system with trees is possible	dimensionless	0 or 1 (1)	RUN & OUTPUT SECTION
2.	AF_Circ?	Switch to decide on circular versus linear symmetry. 1 = circular system, 0 = linear system	dimensionless	0 or 1 (0)	(AF System)
3.	AF_Crop?	Parameter governing an option to simulate system with crop. Value 0 means system without crop, value 1 means system with crop	dimensionless	0 or 1 (0)	RUN & OUTPUT SECTION
4.	AF_DeepSubSoil	Equivalent depth of the subsoil below layer 4, that is used to calculate the effective water outflow from the soil column, via $S_{KsatVDeepSub}$	m	0 - 10 (3)	Agroforestry Zone
5.	AF_DepthDynamic?	Switch for making the depth of soil layer 1 on sloping land system a dynamic property	dimensionless	0 or 1 (0)	Agroforestry Zone/Sloping Land and Parkland System
6.	AF_DepthGroundWater Table	Depth of groundwater table below the bottom of layer 4, expressed in m. For the time being the value is used as a constant in defining 'field capacity'.	m	0 - 10 (0)	Agroforestry Zone
7.	AF_DepthLay[Zone]	Soil depth increment in (= layer thickness of) i -th soil layer, $i = 1, 2, 3, 4$. For sloping land systems the value for the layer 1 is used as average topsoil depth at the start of the run; actual depth of layer 1 will be calculated from the two AF_Slope parameters	m	0 - 1 (.05, .15, .3, .5 for $i = 1, \dots, 4$)	(AF System)
8.	AF_DynPestImpacts?	Switch governing an option to simulate system with dynamic pest impact. Value 0 means no dynamic pest impacts, value 1 means dynamic pest impacts is possible.	dimensionless	0 - 1 (0)	RUN & OUTPUT SECTION
9.	AF_PlotNumberUphill	Number of similar uphill plot neighbors as source of Lateral Inflow & Run-on	dimensionless	(0)	Agroforestry zone
10.	AF_RunNutLim[SoilNut]?	Switch governing an option to simulate system with nutrient limitation. Value 0 means no nutrient limitation, value 1 means nutrient is possible.	dimensionless	0 - 1 (1)	RUN & OUTPUT SECTION
11.	AF_RunOnFrac	Fraction of surface runoff from the area uphill that enters the simulation area as run-on.	dimensionless	0 - 1 (0)	Agroforestry Zone
12.	AF_RunWatLim?	Parameter governing an option to simulate system with water limitation. Value 0 means no water limitation, value 1 means water limitation is possible.	dimensionless	0 - 1 (1)	RUN & OUTPUT SECTION

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
13.	AF_SimulateWeeds?	Parameter governing an option to simulate weed growth. Value 0 means no weed growth, value 1 means weed will start growing whenever crop is absent.	%	0 or 1 (0)	RUN & OUTPUT SECTION
14.	AF_SlopeSurfInit	Slope (expressed as percent elevation increment per horizontal distance) of the soil surface at the start of the simulation; this value can differ from the slope of the soil profile AF_SlopeSoil-Horiz, but should not differ too much.	%	0 – 100 (0)	Agroforestry Zone/Sloping Land and Parkland System
15.	AF_SlopeSoilHoriz	Slope (expressed as percent elevation increment per horizontal distance) of the soil horizons below the surface, especially that of the topsoil, used to calculate actual topsoil depth per zone.	%	0 – 100 (0)	Agroforestry Zone/Sloping Land and Parkland System
16.	AF_StoneFrac[Zone,SoilLayer]	Fraction of stone in each soil layer and zone	dimensionless	0 – 1 (0)	Agroforestry Zone
17.	AF_TreePosit[Tree]	Position of each tree type. It can be in zone 1 (1) or zone 4 (4); if one wants it to be in both, two otherwise equal tree types can be defined.	dimensionless	1 or 4 (1)	(AF System)
18.	AF_WeedZn[Zone]	Switch 0 or 1 to have weeding application for each zone (1 = weeding application, 0 = no weeding)	dimensionless	0 or 1 (0)	Management/Weed Growth
19.	AF_Zone[Zone]	Width of each zone. Width of zone 4 is calculated back from AF_ZoneTot minus the sum of zone 1+2+3	m	0 – 100 (.5, 1,1)	(AF System)
20.	AF_ZoneTot	Total width of agroforestry system simulated	m	0 – 100 (3.5)	(AF System)
21.	C_AgronYMoistFrac[Cr]	Standard moisture content for expressing marketable yields of each crop	dimensionless	0 – 1 (.15)	(Crop Library)
22.	C_ApplyMaintResp?	On/Off switch for applying the maintenance respiration; 1 = on, 0 = off	dimensionless	0 or 1 (0)	Maintenance Respiration
23.	C_DailyWeedSeedDecayFrac	Fraction of the weed seed bank that loses viability and is transferred to the litter pool for decomposition	fraction day ⁻¹	0 – 1 (.02)	Management/Weed Growth
24.	C_HostEffForT1[Cr]	Effectiveness of crop roots as host for a parasitic tree (T1)	cm ³ cm ⁻¹	(0)	Root Parasitism
25.	C_RelRespGroRes	Relative weighting factor for growth used in calculating daily maintenance respiration	dimensionless	(.5)	Maintenance Respiration
26.	C_RelRespRt	Relative weighting factor for roots used in calculating daily maintenance respiration	dimensionless	(.3)	Maintenance Respiration
27.	C_RelRespSLV	Relative weighting factor for stem & leaves used in calculating daily maintenance respiration	dimensionless	(.5)	Maintenance Respiration

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
28.	C_RelRespYieldCurr	Relative weighting factor for developing fruits/Yield as part of total biomass as used for maintenance respiration	dimensionless	(1)	Maintenance Respiration
29.	C_ResidRemovalFrac	Fraction of crop residue removed from field (not returned as mulch). The same value applies for all zones and all crops used in the simulation	fraction	0 – 1 (0)	Management/Mulching
30.	C_RespperBiom	The relative use of resources for maintenance respiration per unit biomass	dimensionless	0 – 0.2 (.05)	Maintenance Respiration
31.	C_RespTemp	A graphical relation between temperature and maintenance respiration	dimensionless	(see C_RespTemp graph)	Maintenance Respiration
32.	C_Tmin	Minimum Air Temperature for Crop	0 Celsius	20	Temperature
33.	C_TOpt	Optimum Air Temperature for Crop	0 Celsius	21	Temperature
34.	C_WeedGermFrac	Fraction of weed seeds in the seed bank that germinates when a new opportunity arises, e.g. at the end of a cropping season	fraction	0 – 1 (.1)	Management/Weed Growth
35.	C_WeedSeedBanknit	Initial dry weight of weed seeds in seed bank	kg m ⁻²	0 – 1 (.01)	Management/Weed Growth
36.	C_WeedSeedExtInflux	Daily influx of weed seeds from outside of the plot	kg m ⁻² day ⁻¹	0 – 0.1 (.00001)	Management/Weed Growth
37.	Ca_CType[Zone]	A graphical input parameter governing the type of crop planted in sequence, with the possibility of having different crops (and/or planting times) in different zones. Associated with type of crop in database. See Wanulcas.xls	dimensionless	1 – 5 (2)	(Crop Management)
38.	Ca_DoYStart	Day of year at which simulation starts	julian days	1 – 365 (300)	RUN & OUTPUT SECTION
39.	Ca_ExtOrgApp?[Type]	Parameter governing an option to have simulation with applying external organic input or not. Value 0 means not applying external organic input, value 1 means applying external organic input	dimensionless	0 or 1 (0)	(Crop Management)
40.	Ca_FertApp?[SINut]	Parameter governing an option to have simulation with applying fertilizer or not. Value 0 means not applying fertilizer, value 1 means applying fertilizer	dimensionless	0 or 1 (1)	(Crop Management)
41.	Ca_FertOrExtOrgApp-Amount[Zone]	Amount of N or P fertilizer or external organic applied. A graphical input parameter.	g m ⁻²	0 – 10 (4.5 for each N and P)	(Crop Management)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
42.	Ca_FertOrExtOrgAppDoY[SiNut]	Time of fertilizer or external organic input application. A graphical input parameter.	julian days	1 – 365 (see excel sheet Crop Management)	(Crop Management)
43.	Ca_FertOrExtOrgAppYear[SiNut]	Year of fertilizer or external organic input application. A graphical input parameter.	dimensionless	any integer value (see excel sheet Crop Management)	(Crop Management)
44.	Ca_ImmAmount[P,Zone]	Amount of immobile P fertilizer applied. A graphical input parameter	g m ⁻²	(see Ca_ImmAmount graph)	Management/P Immobile Input
45.	Ca_ImmDOY[P]	Time of immobile P fertilizer application. A graphical input parameter	julian days	1 – 365 (see Ca_ImmDOY graph)	Management/P Immobile Input
46.	Ca_ImmY[P]	Year of immobile P fertilizer application. A graphical input parameter	dimensionless	Any integer value (see Ca_ImmY graph)	Management/P Immobile Input
47.	Ca_PlantDoY[Zone]	Day of crop planting for each subsequent crop. A graphical input parameter.	julian days	1 – 365 (see excel sheet Crop Management)	(Crop Management)
48.	Ca_PlantYear[Zone]	Year of planting for each subsequent crop. A graphical input parameter	dimensionless	any integer value (see excel sheet Crop Management)	(Crop Management)
49.	Cq_CHarvAlloc[Cr]	Allocation of biomass to harvested parts (grain, tuber) as a function of crop growth stage.	dimensionless	(see Cq_CHarvAlloc table)	(Crop Library)
50.	Cq_ClosedCan[Cr]	Amount of crop canopy biomass at which canopy is closed and nutrient demand per unit new biomass shifts from Cq_ConcY-Young to Cq_ConcOld.	kg m ⁻²	0 – 0.5 (0.2)	(Crop Library/Nutrient Uptake)
51.	Cq_CLWR[Cr]	Crop leaf weight ratio = gram of green leaf area per gram of shoot, for each crop species as a function of crop growth stage.	g m ⁻²	(see Cq_CLWR table)	(Crop Library)
52.	Cq_ConcOld[Cr, SiNut]	Nutrient concentration in crop tissue formed after biomass has reached the Cq_ClosedCan value.	dimensionless	0 – 0.1 (N = .01, P = .0025)	(Crop Library/Nutrient Uptake)
53.	Cq_ConcRt[Cr]	N concentration in crop roots	dimensionless	0 – 0.1 (.01)	(Crop Library/Nutrient Uptake)
54.	Cq_ConcYoung[Cr, SiNut]	Nutrient concentration in young crop biomass (before biomass has reached the Cq_ClosedCan value).	dimensionless	0 – 0.1 (N = .015 P = .007)	(Crop Library/Nutrient Uptake)
55.	Cq_CovEff[Cr]	Crop Cover Efficiency factor, used in calculating erosion (Erosion type 1)	dimensionless	0 – 1 (1)	(Crop Library/Soil Erosion)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
56.	Cq_CRelLUE[Cr]	Relative light use efficiency (fraction of Cq_GroMax achieved per unit light capture) for each type of crop grown as a function of crop growth stage.	dimensionless	(see Cq_CRelLUE table)	(Crop Library)
57.	Cq_DOYFlwBeg[Cr]	The earliest day in a year when crop start to flowers	julian days	1 – 365 (1)	(Crop Library/Annual or Perennial?)
58.	Cq_DOYFlwEnd[Cr]	The latest day in a year when crop start to flowers	julian days	1 – 365 (365)	(Crop Library/Annual or Perennial?)
59.	Cq_GroMax[Cr]	Maximum daily dry matter production rate at full light capture, for each crop species under local conditions	kg m ⁻² day ⁻¹	0.001 – 0.1 (.014)	(Crop Library/Crop Growth)
60.	Cq_Gseed[Cr]	Seed weight (initial C_CarbHydReserves to be used for growth).	kg m ⁻²	0.001 – 0.1 (.004)	(Crop Library/Crop Growth)
61.	Cq_HBiomConv[Cr]	Factor for conversion of crop biomass increment (up to crop stage 1) to crop height increment	dimensionless	0.1 – 10 (7)	(Crop Library/Crop Growth)
62.	Cq_kLight[Cr]	Light extinction coefficient for the crop canopy = efficiency of crop foliage in absorbing light.	dimensionless	0 – 1 (.65)	(Crop Library/Light Capture)
63.	Cq_LAI_Max	Maximum leaf area index for the crop; if more biomass is produced a proportional amount is transferred to the litter layer	dimensionless	(5)	(Crop Library/Canopy)
64.	Cq_LignResid[Cr]	Lignin concentration of crop residue (eg. 20%=0.2).	dimensionless	0 – 1 (.2)	(Crop Library/Litter Quality)
65.	Cq_LignRootRes[Cr]	Lignin concentration of crop root residues	dimensionless	0 – 1 (.2)	(Crop Library/Litter Quality)
66.	Cq_Lp[Cr]	Hydraulic conductivity of crop roots, reflecting the physiological entry resistance to water per unit root length and unit gradient.	cm day ⁻¹	0 – 0.00001 (.00001)	(Crop Library/Water Uptake)
67.	Cq_MaxRemob[Cr]	Maximum proportion of stem and leaves remobilized per day to the CarbHydReserves pool, from which it can, for example, be used for growth of the storage component (grain, tuber)	day ⁻¹	0 – 0.1 (.05)	(Crop Library/Crop Growth)
68.	Cq_MycMaxInf[Cr]	Fraction of crop roots infected by mycorrhiza for a soil layer where the Rt_MTInFrac parameter is 1	dimensionless	0 – 1 (.25)	(Crop Library/ Mycorrhiza Fraction)
69.	Cq_NFixDayFrac[Cr]	Fraction of current N deficit derived from atmospheric N ₂ fixation per day for each crop type, if Cq_NFixVariable = 0 ('false').	day ⁻¹	0 – 1 (0)	(Crop Library/N Fixation)
70.	Cq_NFixDWMMaxFrac[Cr]	Maximum fraction of the C_GroRes[Dwl] pool that can be respired for N ₂ fixation if Cq_NFixVariable = 0 ('false')	day ⁻¹	0 – 0.5 (.1)	(Crop Library/N Fixation)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
71.	Cq_NFixDWUnitCost[Cr]	Dry weight cost for respiration per unit N_2 fixation, if Cq_NFixVariable = 0 ('false')	kg [dw] g^{-1} [N]	0 – 1 (0.1)	(Crop Library/N Fixation)
72.	Cq_NFixResp[Cr]	Responsiveness of N_2 fixation to N stress (N in biomass divided by N target), if Cq_NFixVariable = 0 ('false')	dimensionless	0 – 5 (1)	(Crop Library/N Fixation)
73.	Cq_NFixVariable?[Cr]	Switch (0 = false, 1 = true) to choose between variable (N-stress dependent) versus constant N_2 fixation as fraction of N deficit	dimensionless	0 or 1 (0)	(Crop Library/N Fixation)
74.	Cq_NutMobC[Cr, SiNut]	Relative rate of transfer, per unit root length density ($cm\ cm^{-3}$), from the 'immobile' pool of nutrients to the 'mobile' or sorbed pool, due to Crop root activity	$m^2\ day^{-1}$	0 – 0.02 (0)	Crop Library/Crop effect on nutrient mobility
75.	Cq_PotSuctAlphMax[Cr]	Plant potential where transpiration is $(1 - \text{Alpha}) \times \text{potential transpiration}$, Alpha is a small value (e.g. 0.01). Value could be different depend on crop type.	cm	-6000 – -4000 (-5000)	(Crop Library/Water Uptake)
76.	Cq_PotSuctAlphMin[Cr]	Plant potential where transpiration is Alpha * potential transpiration, Alpha is a small value (e.g. 0.01). Value could be different depend on crop type.	cm	-16000 – -14000 (-15000)	(Crop Library/Water Uptake)
77.	Cq_RainWStorCap[Cr]	Rainfall water stored as thin film at leaf surface	mm	0 – 2 (1)	(Crop Library/Rain Interception)
78.	Cq_RelLightMaxGr[Cr]	Relative light intensity at which shading starts to affect tree growth	dimensionless	0 – 1 (1)	(Crop Library/Light Capture)
79.	Cq_RhizEffKaPC[Cr]	Proportional reduction of the apparent adsorption constant for P due to root activity of the crop, expressed as fraction of N_{KaPdef} per unit crop root length density	$m^2\ day^{-1}$	0 – 0.2 (0)	(Crop Library/Crop effect on nutrient mobility)
80.	Cq_RtAlloc[Cr]	Fraction of crop growth reserves allocated to root biomass in the absence of water or nutrient stress as a function of crop stage (only for Rt_Actype=2).	day^{-1}	(see Cq_RtAlloc table)	(Crop Library)
81.	Cq_RtAllocResp[Cr]	Crop root allocation responsiveness to water or nutrient (the factor currently in minimum supply) stress; 0 = constant root allocation, 1 = linear response to water and nitrogen stress, >1 more-than-proportional response (only for Rt_Actype = 2)	dimensionless	0 – 2 (2)	(Crop Library/Roots)
82.	Cq_RtDiam[Cr]	Crop root diameter. It is used in calculating water and nutrient uptake.	cm	0.05 – 1 (.02)	(Crop Library/Roots)
83.	Cq_SingleCycle?[Cr]	A parameter deciding what happens after fruits are ripe: 1 = annual that dies back, 0 = perennial that returns to crop stage = 1.	dimensionless	0 or 1 (1)	(Crop Library/Annual or Perennial?)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
84.	Cq_SLA[Cr]	Crop specific leaf area = green surface area (one-sided) per unit leaf dry weight, for each crop species as a function of crop growth stage. For Cq_Atype =1, ..., 5, default values are provided. Cq_Atype = 6, ...,10 user defined, as before.	m ² g ⁻¹	(see Cq_SLA table)	(Crop Library)
85.	Cq_TimeGen[Cr]	Length of generative stage for each crop. For Cq_Atype =1, ..., 5, default values are provided, but can be modified to adopt the default crop parameters to local conditions.	days	0 – 1000 (30)	(Crop Library/Crop Stage)
86.	Cq_TimeVeg[Cr]	Length of vegetative stage for each crop. For Cq_Atype =1, ..., 5, default values are provided, but can be modified to adopt the default crop parameters to local conditions.	days	0 – 1000 (60)	(Crop Library/Crop Stage)
87.	Cq_TranspRatio[Cr]	Amount of water needed per unit dry matter production of each crop species. For Cq_Atype =1, ..., 5, default values are provided. For Cq_Atype=6, ..., 10 user defined	l kg ⁻¹	200 – 600 (300)	(Crop Library/Crop Growth)
88.	Cq_WeedType	Weed type. This is user defined. Weed biomass growth follows the rules of crop growth. It takes the same type of parameters as crop. All the related input parameters are in Excel sheet	dimensionless	(5)	Management/Weed Growth
89.	CW_EnergyDrivenEpot?	Switch (1 = yes, 0 = no) to determine whether the crop water demand driven by Epot	dimensionless	(1)	Soil Evaporation
90.	E_BulkDens	Bulk density used in converting soil mass movement to changes in volume of topsoil per zone	g cm ⁻³	0.5 – 1.6 (1.4)	(Soil Hydraulic)
91.	E_CovEffT[Tree]	Tree cover efficiency factor (per unit tree LAI)	dimensionless	0 – 1 (.5)	(Tree Library/Erosion Protection)
92.	E_EntrallmentCoeffBare-Plot	Entrallment coefficient for sediment movement (Rose equation) In the absence of vegetative soil cover	Ton (soil) mm ⁻¹ m ²	0 – 1 (.002)	Soil Erosion and Sedimentation
93.	E_ErosiType	Parameter to decide on model of erosion used. 1 = using USLE, 0 = using Rose Equation	dimensionless	0 or 1 (0)	Soil Erosion and Sedimentation
94.	E_IntvPloughPlant	Length of ploughing time	Julian days	1 – 365 (10)	Management/Tillage
95.	E_PloughBefPlant?	Paramater governing option to plough before planting (0 = no ploughing, 1 = ploughing before planting)	dimensionless	0 or 1 (0)	Management/Tillage
96.	E_PloughDoY	Date of ploughing	Julian days	1 – 365 (364)	Management/Tillage
97.	E_PloughY	Year of ploughing	dimensionless	0 – 100 (100)	Management/Tillage
98.	E_RainFac	A multiplier determining impact of rainfall on soil erosion, for calculation soil loss using USLE	dimensionless	0 – 10 (1)	Soil Erosion and Sedimentation

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
99.	E_SoilMoveperPlough	Amount of soil moved per ploughing event, for calculation soil loss using USLE	kg m ⁻²	0 – 500 (399)	Soil Erosion and Sedimentation
101.	E_SoilType	Type of soil. 1 = medium, 2 = sandy, 3 = clay	dimensionless	1, 2, 3 (1)	Soil Erosion and Sedimentation
101.	E_TillZone?[Zone]	On/off switch for tilling activity in each zone (0 = no tillage, 1 = with tillage)	dimensionless	0 or 1 (0, 1, 1, 1)	Management/Tillage
102.	Evap_InitSlashM	Initial moisture content of slashed vegetation	fraction	(0.4)	Soil Evaporation
103.	Evap_InitWoodM	Initial moisture content of slashed wood	fraction	(0.25)	Soil Evaporation
104.	Evap_MulchEffSurfLit	Effect of mulch on the amount of water evaporating from the soil	dimensionless	1	Soil Evaporation
105.	Evap_SlashDryFact	Factor determined of water of slashed vegetation will evaporated	fraction	0 – 1 (0.5)	Soil Evaporation
106.	Evap_TransRedFractionByCan InterceptedWater	The evaporation of water intercepted by plant canopies will reduce the potential transpiration by satisfying part of the energy-driven potential evapotranspiration; this parameter determines the fraction of canopy interception transpiration that will be reduce from Epot before we determine plant demand. Lower values would reflect: 1) rainfall at night (evaporation not during peak of transpiration) 2) more open landscapes where more dry wind comes in from outside the plot.		(0.5)	Soil Evaporation
107.	Evap_WoodDryFact	Factor determined of water of slashed wood will evaporated	fraction	0 – 1 (0.25)	Soil Evaporation
108.	GHG_LitMinMultiplier	Multipplier of litter mineralization for quick modifications of nitrogen oxide emission	dimensionless	(1)	Soil Water & Nutrient/Nox emissions
109.	GHG_N ₂ per_NOx	Ratio of nitrous and nitric oxide	dimensionless	See graph GHG_N ₂ perNOx	Soil Water & Nutrient/Nox emissions
110.	LF_FracGWReleaseAsInflow	The fraction of groundwater that flow out that reaches the simulated zone (this depends on subsoil stratification and landscape characteristics beyond the scope of out current model)	dimensionless	(0)	Agroforestry Zone
111.	LF_GWReleaseFraction	The fraction of the current stock of groundwater that flows out on a daily basis. A stock of groundwater stored uphill depends on the 'number of plots uphill'.	dimensionless	(.05)	Agroforestry Zone

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
112.	LF_SubSurfInflow4	Amount of sub surface water inflow in layer 4	mm day ⁻¹	0 – 5 (0)	Agroforestry zone
113.	Mc_Carbon	Proportion of total carbon in plant litter and residue	dimensionless	0 – 0.5 (.42)	Soil Organic Matter and Litter Quality/Litter Quality
114.	Mc_CExtOrg[type]	Carbon concentration of external input	dimensionless	0 – 1 (.4)	Litter Quality/Quality of Ext. Organic Input
115.	Mc_CNRatioMetab[zone]	Initial C:N ratio metabolic pool of litter	dimensionless	(8)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
116.	Mc_LignExtOrg[Type]	Lignin concentration of external input.	dimensionless	0 – 1 (.2)	Soil Organic Matter and Litter Quality/Quality of Ext. Organic Input
117.	Mc_PolypExtOrg[Type]	Polyphenol concentration of external input	dimensionless	0 – 1 (0)	Soil Organic Matter and Litter Quality/Quality of Ext. Organic Input
118.	Mc_RelKActLit	Decomposition rate of active surface litter pool relative to decomposition rate of active soil fraction pool. It's assumed the decomposition rate of active surface litter and active soil fraction is the same.	dimensionless	(.8)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
119.	Mc_RelKMetabLit	Decomposition rate of metabolic surface litter pool relative to decomposition rate of metabolic soil litter pool. It's adopted from century model, the decomposition rate of metabolic surface litter pool = 0.028 per week = 0.04 per day and the decomposition rate of metabolic soil litter pool = 0.35 per week = 0.05 per day.	dimensionless	(.8)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
120.	Mc_RelKPassLit	Decomposition rate of passive surface litter pool relative to decomposition rate of passive soil fraction pool. It's assumed the decomposition rate of passive surface litter and passive soil fraction is the same.	dimensionless	(1)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
121.	Mc_RelKSlwLit	Decomposition rate of slow surface litter pool relative to decomposition rate of slow soil fraction pool. It's assumed the decomposition rate of slow surface litter and slow soil fraction is the same.	dimensionless	(1)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
122.	Mc_RelKStrucLit	Decomposition rate of structural surface litter pool relative to decomposition rate of structural soil litter pool. It's adopted from century model, the decomposition rate of structural surface litter pool = 0.076 per week = 0.010857 per day and the decomposition rate of structural soil litter pool = 0.094 per week = 0.013429 per day.	dimensionless	(0.808511)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
123.	Mc2_Clay	Proportion of clay in soil (only for soil organic matter type 2)	dimensionless	0 – 1 (.316)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
124.	Mc2_ClayCoeffCref	Coefficient of clay based on tabulated Cref for soil organic matter type 2	dimensionless	(.94)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
125.	Mc2_CNRatInitMetab[zone]	Initial C:N ratio metabolic pool of soil organic matter	dimensionless	(8)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
126.	Mc2_CorgInitMeth3	Initial soil organic carbon value in soil organic matter pool using Type 3.	gr cm ⁻²	(2)	Soil Organic Matter and Litter Quality/ Initial C & N in SOM Pool
127.	Mc2_CorgpCref	Initial soil organic carbon value in soil organic matter pool using Type 3.	gr cm ⁻²	(.8)	Soil Organic Matter and Litter Quality/ Initial C & N in SOM Pool
128.	Mc2_CrefMeth3	Initial C-ref value in soil organic matter pool using Type 2.	gr cm ⁻²	(3)	Soil Organic Matter and Litter Quality/ Initial C & N in SOM Pool
129.	Mc2_CrefOffset	Constant for C reference tabulated for soil organic matter type 2.	dimensionless	1.256	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
130.	Mc2_KAct	Decay rate for decomposition of active pool of soil organic matter	dimensionless	(.02)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
131.	Mic2_kMetab	Decay rate for decomposition of active pool of soil organic matter	dimensionless	(.05)	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
132.	Mic2_kPass	Decay rate for decomposition of active pool of soil organic matter	dimensionless	(.0000186)	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
133.	Mic2_kRelLayer[SoilLayer]	Factor determined decay rate for decomposition of active, passive and slow pool for each soil layer (relative to decay rate for decomposition for each pool).	fraction	(1, .8, .7, .6)	Soil Organic Matter and Litter Quality/ Other Factors Affecting Decomposition
134.	Mic2_KSlw	Decay rate for decomposition of active pool of soil organic matter	dimensionless	(.000543)	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
135.	Mic2_KStruc	Decay rate for decomposition of active pool of soil organic matter	dimensionless	(.013429)	Soil Organic Matter and Litter Quality/Other Factors affecting Decomposition
136.	Mic2_pH	Soil pH (only for soil organic matter type 2)	dimensionless	(5)	Soil Organic Matter and Litter quality/ Other Factors Affecting Decomposition
137.	Mic2_pHCoeffCref	Coefficient of pH based on tabulated of Cref for soil organic matter type 2	dimensionless	-.156	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
138.	Mic2_RainTransfer[Pool]	Rain factor which control transferring process of litter to SOM pool	dimensionless	.001	Soil Organic Matter and Litter Quality/Litter → SOM Transfer
139.	Mic2_Silt	Proportion of silt in soil (only for soil organic matter type 2)	dimensionless	0 – 1 (.2)	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
140.	Mic2_SiltClayCoeffCref	Coefficient of clay and silt based on tabulated Cref for soil organic matter type 2	dimensionless	(.703219)	Soil Organic Matter and Litter Quality/Other Factors Affecting Decomposition
141.	Mic2_SoilTillTransfer[Pool]	Soil tillage factor which control transferring process of litter to SOM pool	dimensionless	(1)	Soil Organic Matter and Litter Quality/Litter → SOM Transfer

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
142.	Mc2_SOMDistribution[SoilLayer]	Relative distribution of carbon between different soil layers	fraction	0 – 1 (1, .2, .1, .05)	Soil Organic Matter and Litter Quality/SOM Distribution
143.	Mc2_SOMInitType	Parameter defining methods to initialize soil organic matter pool. Three methods are provided for initializing the soil organic matter pools: Type = 1 the user can specify all pool sizes for all zones, Type = 2 the user can specify the size of all pools relative to those for a forest soil (Cref) that is calculated from soil texture data, Type = 3 the user specifies the Corg and Cref directly, but otherwise follows the procedure of Type 2	dimensionless	1, 2, 3 (1)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
144.	Mc2_WormTransfer[Pool]	Worm factor which control transferring process of litter to SOM pool	dimensionless	0.003 – 0.1 (Struc, Metab and Act = .1; Slow = .3; Pass = .003)	Soil Organic Matter and Litter Quality/ Litter → SOM Transfer
145.	Mn_CNAct	C:N ratio of active pools	dimensionless	5 – 10 (8)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
146.	Mn_CNPass	C:N ratio of passive pools	dimensionless	8 – 15 (11)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
147.	Mn_CNSlow	C:N ratio of slow pools	dimensionless	8 – 15 (11)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
148.	Mn_CNStruc	C:N ratio of structural pools	dimensionless	100 – 200 (150)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
149.	Mn_ExtOrgN[Type,SiNut]	N or P concentration of external input	dimensionless	0 – 0.1 (N = .05, .1; P = .005, .001)	Soil Organic Matter and Litter Quality/Quality of Ext. Organic Input
150.	Mn_FertDissFrac[SiNut]	Daily fraction of fertilizer dissolved	Dimensionless	(N = 0.3, P = 0.5)	Soil Water & Nutrient/Fertilizer Movement



No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
151.	Mn_InitAct[Zone]	Initial amount of N in active Litter pool of each zone	mg cm ⁻³	0 – 1 (.00002)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
152.	Mn_InitMetab[Zone]	Initial amount of N in metabolic Litter pool of each zone	mg cm ⁻³	0 – 1 (0)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
153.	Mn_InitPass[Zone]	Initial amount of N in passive Litter pool of each zone	mg cm ⁻³	0 – 1 (.00001)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
154.	Mn_InitSlw[Zone]	Initial amount of N in slow Litter pool of each zone	mg cm ⁻³	0 – 1 (.000001)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
155.	Mn_InitStruc[Zone]	Initial amount of N in structural Litter pool of each zone	mg cm ⁻³	0 – 1 (0)	Soil Organic Matter and Litter Quality/Initial C & N in Litter Pool
156.	Mn_LatFlowFertKm	Runoff flow that causes half of the (undissolved) surface fertilizer to move to the next zone	mm	(10)	Soil Water & Nutrient/Fertilizer Movement
157.	Mn_NutRatAct[P]	Ratio of N to P (N:P) in active organic matter pools	dimensionless	1 – 10 (10)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
158.	Mn_NutRatMetab[P]	Ratio of N to P (N:P) in metabolic organic matter pools	dimensionless	1 – 10 (10)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
159.	Mn_NutRatPass[P]	Ratio of N to P (N:P) in passive organic matter pools	dimensionless	1 – 10 (10)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
160.	Mn_NutRatSlw[P]	Ratio of N to P (N:P) in slow organic matter pools	dimensionless	1 – 10 (10)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool
161.	Mn_NutRatStruc[P]	Ratio of N to P (N:P) in structural organic matter pools	dimensionless	1 – 10 (10)	Soil Organic Matter and Litter Quality/C:N Ratio of Litter Pool

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
162.	Mn2_InitAct[Zone]	Initial amount of N in active SOM pool of each zone	mg cm ⁻³	0 – 1 (.2)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
163.	Mn2_InitMetab[Zone]	Initial amount of N in metabolic SOM pool of each zone	mg cm ⁻³	0 – 1 (0)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
164.	Mn2_InitPass[Zone]	Initial amount of N in passive SOM pool of each zone	mg cm ⁻³	0 – 1 (3.9)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
165.	Mn2_InitSlw[Zone]	Initial amount of N in slow SOM pool of each zone	mg cm ⁻³	0 – 1 (1)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
166.	Mn2_InitStruc[Zone]	Initial amount of N in structural SOM pool of each zone	mg cm ⁻³	0 – 1 (0)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
167.	Mn2_PassRelLayer[SoilLayer]	Proportional distribution of passive pool	fraction	(1, 1.2, 1.4, 1.6)	Soil Organic Matter and Litter Quality/Initial C & N in SOM Pool
168.	N_BypassMacro[Zone]	Preferential flows of nutrients in the leachate relative to average concentration * water flow; values < 1 indicates retardation of nutrients due to bypass flow of water in macropores at soil layer <i>i</i>	dimensionless	0 – 2 (1)	Soil Water and Nutrient/ Macropore Bypass Flow
169.	N_DiffCoef[SiNut]	Nitrogen diffusion coefficient	cm ² day ⁻¹	0 - 1 (N = 1, P = .76896)	Soil Water and Nutrient /Diffusivity coefficient
170.	N_FracNO3[Zone]	Fraction of NO ₃ of total N in <i>i</i> -th soil layer	dimensionless	0 - 1 (0.4)	Soil Water and Nutrient/ Nitrate Fraction
171.	N_ImInit[Zone, SiNut]	Initial amount of nutrient in immobile pool of each zone	mg cm ⁻³	0 – 0.1 (N = .05, P = .01)	Soil Water and Nutrient/ Initial Immobile Nutrient
172.	N_KaNH4[Zone]	Apparent (instantaneous) adsorption constant or ratio of amount NH ₄ adsorbed and amount in solution for <i>i</i> -th layer	mg cm ⁻³	0 – 1 (5)	Soil Water and Nutrient/Adsorption constant for N

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
173.	N_KaNO3[Zone]	Apparent (instantaneous) adsorption constant or ratio of amount NO ₃ adsorbed and amount in solution for i-th layer	mg cm ⁻³	0 – 1 (.3)	Soil Water and Nutrient /Adsorption constant for N
174.	N_KaPDefi[Zone]	Apparent (instantaneous) adsorption constant for inorganic P, or ratio of amount of inorganic P adsorbed ant the amount in soil solution; the adsorption constant depends on the P concentration on soil solution and is read in a tabular form (as graphical input parameter).	mg cm ⁻³	(see N_KaPDefi table)	(Phosphorus/to P sorption data)
175.	N_Lat4InflowRelConc	Nutrient concentrations in the incoming sub-surface flows into zone 4, relative to the current average nutrient concentration in that layer across all zones in the simulated area	dimensionless	0 – 10 (1)	Agroforestry Zone
176.	N_Nlnti[Zone, SiNut]	Initial amount of nutrient in soil layer i of each zone	mg cm ³	0 – 0.5 (N layer 1 = .003, layer 2 – 4 = .01 ; P layer 1 = .1, layer 2 = .08, layer 3 – 4 = .04)	For P, (Phosphorus/Initial P availability Index per Zone and Layer) For N, (Nitrogen)
177.	N_NutMob[SiNut]	Relative rate of transfer from the 'immobile' pool of nutrients to the 'mobile' or sorbed pool, due to processes other than root activity in soil layer i	day ⁻¹	0 – 0.02 (0)	Soil Nutrient/Nutrient Mobilization
178.	N_RtSynlocl	Root synlocation, or degree to which roots of the crop and tree are co-occurring within the various soil layers, affecting the way in which benefits of rhizosphere modification are shared; 1 = sharing of rhizosphere modifications by all roots present, based on their share in total root length, 0 = complete monopoly by roots modifying the rhizosphere	dimensionless	0 – 1 (.5)	Roots and Mycorrhiza
179.	N_UseNGassLossEst?	A switch determining simulate system with gaseous N losses. 0 = no gaseous N losses, 1 = with gaseous N losses	dimensionless	(0)	Soil Water & Nutrient
180.	N15_Addi [Zone]	Initial amount of N ¹⁵ in soil layer i of each zone	mg cm ⁻³	(0)	N ¹⁵ model sector
181.	P_BurnLab	Amount of labour involved in burning the field per unit simulated filed	person days ha ⁻¹	(see excel sheet Profitability)	(Profitability)
182.	P_CferrPrice[SiNut, Price]	Cost of fertilizer at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Profitability)	(Profitability)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
183.	P_CharvLab[Cr]	Amount of labour involved in harvesting crop products per unit dry weight	person days Mg ⁻¹ per cropping season	(see excel sheet Crop Library)	(Crop Library/Profitability)
184.	P_CNuFerAppperCropSeason	Number of fertilizer application per cropping season	dimensionless	Any integer value (2)	Profitability
185.	P_CPestContLab[Cr]	Amount of labour involved in pest control per cropping season	person days ha ⁻¹ per cropping season	(see excel sheet Crop Library)	(Crop Library/Profitability)
186.	P_CPestContPrice[Price]	Amount of direct costs (outside labour) involved in pest control per cropping season	currency unit per ha ⁻¹ per cropping season	(see excel sheet Profitability)	(Profitability)
187.	P_CPlantLab[Cr]	Amount of labour involved in planting per cropping season	person days ha ⁻¹ per cropping season	(see excel sheet Crop Library)	(Crop Library/Profitability)
188.	P_CropProfThreshold	Threshold value for crop profitability. Relevant to parameter P_UseCropStopRule? = 1	Currency unit	(100000)	Management/Ending a Crop Cycle
189.	P_CSeedPrice[Cr,Price]	Cost of crop seed per kg at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Crop Library)	(Crop Library/Profitability)
190.	P_CWeedLab[Cr]	Amount of labour involved in weeding per cropping season	person days ha ⁻¹ per cropping season	(see excel sheet Crop Library)	(Crop Library/Profitability)
191.	P_CyieldPrice[Cr, Price]	Price of crop yield per unit dry weight at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Crop Library)	(Crop Library/Profitability)
192.	P_DiscountRate	Discount rate (% per year) that applies to both social and private prices	% year ⁻¹	(see excel sheet Profitability)	(Profitability)
193.	P_ExtOrgPrice [Type,Price]	Price of external organic input	currency unit kg ⁻¹	(see excel sheet Profitability)	(Profitability)
194.	P_FenceMatCost[Price]	Price of off-farm material used for building or maintaining a fence around the field	currency unit ha ⁻¹	(see excel sheet Profitability)	(Profitability)
195.	P_LabourforPestContrl ?	A switch 0 or 1 for pesticide application related to the use of labour (1 = pesticide application, 0 = no pesticide application)	dimensionless	0 or 1	Profitability
196.	P_LabourforWeed ?	A switch 0 or 1 for weeding application related to the use of labour (1 = weeding application, 0 = no weeding application)	dimensionless	0 or 1	Profitability

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
197.	P_TFruitHarvLab	Amount of labour involved in harvesting fruits per unit dry weight	person days kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
198.	P_TFruitPrice[Price]	Price of tree fruit yield per unit dry weight at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
199.	P_TLatexHarvLab	Amount of labour involved in harvesting latex per unit dry weight	person days kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
200.	P_TLatexPrice[Price]	Price of tree latex yield per unit dry weight at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
201.	P_TNuFerApp[Tree]	Number of fertilizer application per year	dimensionless	Any integr value (2)	Profitability
202.	P_TPPlantLab	Amount of labour involved in planting trees per unit dry weight	person days kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
203.	P_TPPrunLab[Tree]	Amount of labour involved in pruning trees per unit dry weight	person days kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
204.	P_TPPrunPrice[Price]	Price of tree prunings harvested from the field per unit dry weight at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
205.	P_TSeedPrice[Price]	Costs of tree planting material per unit initial tree biomass at social and private prices, respectively.	currency unit tree ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
206.	P_TWWoodHarvLab	Amount of labour involved in harvesting wood products per unit dry weight	person days kg=1	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
207.	P_TWWoodPrice[Price]	Price of tree wood product yield per unit dry weight at social and private prices, respectively.	currency unit kg ⁻¹	(see excel sheet Tree Library)	(Tree Library/Profit- ability)
208.	P_UnitLabCost[Price]	Cost per unit labour at social and private prices, respectively	currency unit person days ⁻¹	(see excel sheet Profitability)	(Profitability)
209.	P_UseCropStopRule?	A switch determining the simulation will continue to growth crop or not when the previous crop profitability lower than the threshold value. 0 means continue to growth crop, 1 = stop to growth crop	dimensionless	(0)	Management/Ending a Crop Cycle
210.	PD_CeatenBy[Cr,Animals]	Fraction of crop component lost if eaten by animals. Default animals are pigs, monkey, locust, nematode, goat, buffalo and birds	dimensionless	0 – 1 (0)	(Crop Library/Sensitivity to Pest Damage)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
211.	PD_CFrugivore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a crop frugivore, 1 = animal is frugivore	dimensionless	0 or 1 (0)	Pest and Disease
212.	PD_CFrugivory[Cr]	Constant daily fraction of crop fruit biomass removed due to the action of frugivores	dimensionless	0 – 1 (0)	Pest and Diseases
213.	PD_CHerbivore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a crop herbivore, 1 = animal is herbivore	dimensionless	0 or 1 (0)	Pest and Disease
214.	PD_CHerbivory[Cr]	Constant daily fraction of crop leaf biomass removed due to the action of herbivores	dimensionless	0 – 1 (0)	Pest and Diseases
215.	PD_CRhizovore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a crop rhizovore, 1 = animal is rhizovore	dimensionless	0 or 1 (0)	Pest and Disease
216.	PD_CRhizovory[Cr]	Constant daily fraction of crop root biomass removed due to the action of rhizovores	dimensionless	0 – 1 (0)	Pest and Diseases
217.	PD_FenceBuildDOY	Schedule for day of fencing for each fencing event. A graphical input.	julian days	(see PD_Fence-BuildDOY BuildLab graph)	Pest and Disease
218.	PD_FenceBuildLab	Amount of labour needed to build fence for each fencing event. A graphical input.	person days	(see PD_Fence-BuildLab graph)	Pest and Disease
219.	PD_FenceBuildY	Schedule for year of fencing for each fencing event A graphical input.	dimensionless	(see PD_Fence-BuildY graph)	Pest and Disease
220.	PD_FenceDeck	Daily fractional decay of fence quality	day ⁻¹	0 – 1 (.02)	Pest and Diseases
221.	PD_FenceFullQua	Maximum quality of fence	dimensionless	1 – 4 (2)	Pest and Diseases
222.	PD_FenceMaint?	Switch determining fence maintenance. 1 = fence maintenance will be done automatically, 0 = no fence maintenance	dimensionless	0 or 1 (0)	Pest and Disease
223.	PD_FenceMUnit	Unit improvement of fence quality once it falls below the threshold set in PD_FenceQThresh	dimensionless	0 – 2 (.25)	Pest and Disease
224.	PD_FenceQThresh	Threshold of (relative) fence quality below which labour will be used to repair the fence	dimensionless	0 – 2 (1.1)	Pests and Disease
225.	PD_HalfFenceTime	Time constant of decay of fence quality: time interval after which quality is reduced by 50%	days	0 – 365 (50)	Pest and Disease

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
226.	PD_ JumptheFence?[animals]	The degree to which animals are deterred by a fence from entering the plot	-	0 – 1 (0)	Pest and Diseases
227.	PD_ PopDensOutside[animals]	Population density outside the plot, influencing the presence	-	0 or 1	Pest and Diseases
228.	PD_ TEatenBy?[Animals]	A switch determining tree attacks by specific animals. Default animals are pigs, monkey, locust, nematode, goat, buffalo and birds. 0 = no attack, 1 = attacked	0 or 1 (0)	0 – 1 (0)	(Tree parameters/Pest Impacts)
229.	PD_ TFrugivore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a tree frugivore, 1 = animal is frugivore	dimensionless	0 or 1 (0)	Pest and Disease
230.	PD_ TFrugivory&Abort[Tree]	Constant daily fraction of tree fruit biomass removed due to the action of frugivores	-	0 – 1 (0)	Pest and Diseases
231.	PD_ THerbivore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a tree herbivore, 1 = animal is herbivore	dimensionless	0 or 1 (0)	Pest and Disease
232.	PD_ THerbivory[Tree]	Constant daily fraction of tree leaf biomass removed due to the action of herbivores	-	0 – 1 (0)	Pest and Diseases
233.	PD_ TLignivory[Tree]	Constant daily fraction of tree woody stem biomass removed due to the action of lignivores	-	0 – 1 (0)	Pest and Diseases
234.	PD_ TLignovore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a tree lignovore, 1 = animal is lignovore	dimensionless	0 or 1 (0)	Pest and Disease
235.	PD_ TRRhizovore?[Animals]	A switch determining the presence of attack by each default animal. 0 = animals is not a tree rhizovore, 1 = animal is rhizovore	dimensionless	0 or 1 (0)	Pest and Disease
236.	PD_ TRRhizovory[Tree]	Constant daily fraction of tree root biomass removed due to the action of rhizovores	-	0 – 1 (0)	Pest and Diseases
237.	Rain_ AType	A number 1, 2 or 3 to decide rainfall rate (1= rainfall rate follows precipitation data from external file, rainfall rate follows tabulated data, 2 = rainfall rate follows random generator, 3= rainfall rate follows tabulated monthly total data)	dimensionless	1, 2 or 3 (1)	Rainfall
238.	Rain_ BoundHealI	Boundary value between heavy and light rain (only for Rain_ AType=1)	mm	20 – 30 (25)	Rainfall
239.	Rain_ CoeffVar2	Coefficient variation of rainfall in mm, used in rainfall generated randomly (Rain_AType=2)	dimensionless	0 – 1 (.05)	Rainfall
240.	Rain_ CoeffVar3	Coefficient variation of rainfall in mm, rainfall based on tabulated monthly rainfall (Rain_AType=3)	dimensionless	0 – 1 (.05)	Rainfall

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
241.	Rain_Cycle?	Parameter governing ways to read rainfall data. Corresponds to Rain_AType=1 (0 = use multiple year rainfall data, 1 = use 1 year data in cycle/continuously)	dimensionless	0 or 1 (1)	Rainfall
242.	Rain_Data	Actual daily rainfall data. Entered as graphical function or read from Wanulcas.xls (Stella non-CRT users only). Corresponds to Rain_AType=1.	mm	(see table in excel sheet weather)	(WEATHER)
243.	Rain_DayP	Probability of raining each day as a function of Julian day scaled monthly. Corresponds to Rain_AType=2 and 3.	dimensionless	0 – 1 (.32)	Rainfall
244.	Rain_GenSeed	Seed Random Generator. For Rain_AType=2 and 3.	dimensionless	1 – 32767 (300)	Rainfall
245.	Rain_Heavy	Average precipitation rate of on a heavy rain day; for Rain_AType=2.	mm day ⁻¹	0 – 100 (42)	Rainfall
246.	Rain_HeavyP	Probability of heavy rain; for Rain_AType=2.	dimensionless	0 – 1 (.5)	Rainfall
247.	Rain_IntensCoefVar	Coefficient variance of rain intensity. Rain intensity is a factor affecting water infiltration. It is assumed to follow normal distribution with an average of Rain_IntensMean and standard deviation Rain_IntensMean*Rain_IntensCoefVar.	dimensionless	(.3)	Rainfall
248.	Rain_IntensMean	Average rain intensity per hour. Rain intensity is a factor affecting water infiltration. It is assumed to follow normal distribution with an average of Rain_IntensMean and standard deviation Rain_IntensMean*Rain_IntensCoefVar	mm hr ⁻¹	(50)	Rainfall
249.	Rain_IntercDripRt	The rate of water dripping from water on interception surface	mm hr ⁻¹	(10)	Rainfall
250.	Rain_IntMult	Indicates the maximum temporary storage of water on interception surfaces	dimensionless	(3)	Rainfall
251.	Rain_Light	Average precipitation rate of a light rain day day; for Rain_AType=2.	mm day ⁻¹	0 – 40 (9)	Rainfall
252.	Rain_MaxIntDripDur	Maximum value of water interception delay before start to dripping	mm hr ⁻¹	(.5)	Rainfall
253.	Rain_MonthTot	Tabulated data of monthly rainfall; for Rain_AType=3. Entered as graphical function or read from Wanulcas.xls (Stella non-CRT users only).	mm month ⁻¹	(see Rain_MonthTot graph)	Rainfall
254.	Rain_Multiplier	Multiplier of rainfall for quick modifications of rainfall amount	dimensionless	0 – 4 (1)	RUN & OUTPUT SECTION

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
255.	Rain_PondFlwRt	The rate at which water ponding on the surface will actually flow to a neighbouring zone or plot	mm hr ⁻¹ per m of zone width	(10)	Rainfall
256.	Rain_PondStoreCp	The storage capacity of water ponding on the surface	mm	(5)	Rainfall
257.	Rain_Weight[Zone]	Input weight value to decide amount of rain falling on each zone relative to other zones (eg. equal rainfall in each zone on area basis means 1:1:1:1)	dimensionless	0 – 10 (1)	Rainfall
258.	Rain_YearStart	Initial year based on rainfall data at which simulation starts	dimensionless	any integer value (0)	Rainfall
259.	Rain_UniorBimodal?	An option to have one or two season of rainfall	dimensionless	2	Rainfall
260.	Rain_Probability	Ratio between season 1 and season 2	dimensionless	0.5	Rainfall
261.	Rain_OffsetValue	Influence value of dry month. High value indicates low amount of rainfall at dry season.	dimensionless	-0.5	Rainfall
262.	Rain_ShapeMax	Maximum value of shape. Shape is a basic pattern of rainfall simulator model that use SINUS as the distribution function. This value will be used to determine Pattern value.	dimensionless	1.5	Rainfall
263.	Rain_ShapeMin	Minimum value of shape.	dimensionless	-0.5	Rainfall
264.	Rain_Pattern1Max	Maximum value of pattern. Pattern is calculated from shape value which is the maximum and the minimum value have been adjusted to the maximum and minimum value of total monthly rainfall.	dimensionless	0.06	Rainfall
265.	Rain_Pattern1Min	Minimum value of pattern.	dimensionless	-0.01	Rainfall
266.	Rain_WettestMonthSeason1	The wettest month of the first season (January – June)	dimensionless	1	Rainfall
267.	Rain_WettestMonthSeason2	The wettest month of the second season (July – December)	dimensionless	7	Rainfall
268.	Rain_PeakinessSeason1	The sharpness of the first peak. High value indicates sharper peak.	dimensionless	1	Rainfall
269.	Rain_PeakinessSeason2	The sharpness of the second peak. High value indicates sharper peak.	dimensionless	12	Rainfall
270.	Rain_WeibullParam	Parameter of Weibull distribution. This Weibull parameter is used to predict the value of daily cumulative frequency of rainfall data.	dimensionless	0.93	Rainfall

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
271.	Rain_MonthlyMeanRainfallMax	Maximum value of monthly mean rainfall	mm	333	Rainfall
272.	Rain_MonthlyMeanRainfallMin	Minimum value of monthly mean rainfall	mm	102	Rainfall
273.	Rain_NumberWetDaypM[Month]	The number of wet day for each month	day	See graph	Rainfall
274.	Rain_MonthlyMeanTotal-rainfall [Month]	Monthly mean rainfall on the wet day	mm	See graph	Rainfall
275.	Rain_RelWetPersistencepM [Month]	Monthly relative wet persistence	dimensionless	See graph	Rainfall
276.	Rt_ACType	Parameter governing type of root density data for crop. 0=Lrv data available, 1=Lrv calculated using exponential function model where length root area is constant, 2= Lrv calculated using exponential function model where length root area is derived from root biomass	dimensionless	0, 1, or 2 (0)	Roots and Mycorrhiza/Crop Root
277.	Rt_ATTtype	Parameter governing type of root density data for tree. 0=Lrv data available, 1=Lrv is constant calculated using elliptical function model, 2= Lrv is calculated using elliptical function but dynamically changes according to water or N stress	dimensionless	0, 1 or 2 (0)	Roots and Mycorrhiza/Tree Root
278.	Rt_CDecDepth[Cr]	Parameter governing decrease of crop root with depth; corresponds to Rt_ACType=1 and Cq_ATTtype.	m ⁻¹	0 - 10 (7)	(Crop Library/Roots)
279.	Rt_CDistResp[Cr]	Responsiveness of crop root distribution to the depth at which uptake of the currently limiting resource (water, N or P) is most successful. Value 0 = no response to stress, 0 - 1 = mild response, 1 = proportional change to inverse of relative depth of uptake, > 1 = strong response. Only for Rt_ACType = 2.	dimensionless	0 - 3 (1)	(Crop Library/Roots)
280.	Rt_CHalfLife[Cr]	Crop root half-life (only for Rt_ACType=2)	days	30 - 100 (50)	(Crop Library/Roots)
281.	Rt_CLraConst[Cr]	Total root length per unit area. It is used to calculate crop root density in exponential decrease model (Rt_ACType=1). Also corresponds to Cq_ATTtype.	cm cm ⁻²	0 - 150 (100)	(Crop Library/Roots)
282.	Rt_CLrvmi[Cr]	Maximum crop root length density in i-th soil layer; corresponds to Rt_ACType=0 and Cq_A Type.	cm cm ⁻³	0 - 15 (layer 1 = 5, layer 2 = 3, layer 3 = .3, layer 4 = 0)	(Crop Library/Roots)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
283.	Rt_CMultiplier	Multiplier of root for quick modifications of crop root length density	dimensionless	(1)	Root and Mycorrhiza
284.	Rt_CSRL[Cr]	Specific root length (length per unit dry weight) of crop roots	m g ⁻¹	50 – 100(200)	(Crop Library/Roots)
285.	Rt_MCHypDiam	Diameter of crop mycorrhizal hyphae	cm	0.001 – 0.05 (.01)	Roots & Mycorrhiza/ Mycorrhiza
286.	Rt_MCHypL	Length of crop mycorrhizal hyphae per unit infected root length	dimensionless	10 – 100 (100)	Roots & Mycorrhiza/ Mycorrhiza
287.	Rt_MCInfFraci	Fraction of crop roots that is mycorrhizal (infected) in i-th soil layer	dimensionless	0 – 1 (layer 1 = .05, layer 2 = .25, layer 3 = .05, layer 4 = 0)	Roots & Mycorrhiza/ Mycorrhiza
288.	Rt_MTHypDiam	Diameter of tree mycorrhizal hyphae	cm	0.001 – 0.05 (.01)	Roots & Mycorrhiza/ Mycorrhiza
289.	Rt_MTHypL	Length of tree mycorrhizal hyphae per unit infected root length	dimensionless	10 – 100 (100)	Roots & Mycorrhiza/ Mycorrhiza
290.	Rt_MTInfFraci[Zone]	Fraction of tree roots that is mycorrhizal (infected)	dimensionless	0 – 1 (0)	Roots & Mycorrhiza/ Mycorrhiza
291.	Rt_TAloc[Tree]	Fraction of tree growth reserves allocated to roots in the absence of water or nutrient stress (only for Rt_ATTtype=2)	dimensionless	0 – 1 (.1)	(Tree Library/Roots)
292.	Rt_TAlocResp[Tree]	Responsiveness of tree root allocation to stress factors; 0 = constant root allocation, 1 = linear response to water and nitrogen stress, >1 more-than-proportional response (only for Rt_ATTtype = 2),	dimensionless	0 – 2 (2)	(Tree Library/Roots)
293.	Rt_TDecDepthC[Tree]	Parameter governing decrease of tree root with depth ; for Rt_ATTtype=1	m ⁻¹	0 – 10 (3)	(Tree Library/Roots)
294.	Rt_TDistResp[Tree]	Responsiveness of crop root distribution to the depth at which uptake of the currently limiting resource (water, N or P) is most successful. Value 0 = no response to stress, 0 – 1 = mild response, 1 = proportional change to inverse of relative depth of uptake, > 1 = strong response. Only for Rt_ATTtype = 2.	dimensionless	0 – 5 (2)	(Tree Library/Roots)
295.	Rt_TDistShapeC[Tree]	Tree root distribution shape for Rt_ATTtype=1 and 2; for a value of 1 root length density decreases as much with horizontal as with vertical distance to the tree stem	dimensionless	0 – 2 (.05)	(Tree Library/Roots)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
296.	Rt_THalfLife[Tree]	Tree root half life (only for Rt_ATTtype=2)	days	30 – 150 (60)	(Tree Library/Roots)
297.	Rt_THostEffForT1[Tree]	An option for simulation root parasitism tree 1 on others tree root	dimensionless	(0)	Root Parasitism
298.	Rt_TLengDiam1[Tree]	Length of (branch) roots of a tree root with a proximal (at stem base) diameter of 1 cm; Intercept (a) of allometric equation (RootLength = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	cm cm ^b	0.01 – 1 (10)	(Tree Library/Roots)
299.	Rt_TLengDiamSlope[Tree]	Power coefficient (b) of allometric equation (RootLength = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	1 – 3 (1.5)	(Tree Library/Roots)
300.	Rt_TLraX0[Tree]	Total root length per unit area at X(distance to tree)=0 (tree stem). for Rt_ATTtype=1	cm cm ²	0 – 150 (1)	(Tree Library/Roots)
301.	Rt_TLrvData1[Zone,Tree]	Tree root density in soil layer .i. in each zone; for Rt_ATTtype=0	cm cm ²	0 – 15 (see excel sheet Tree Library/Roots)	(Tree Library/Roots)
302.	Rt_TMultiplier	Multiplier of root for quick modifications of tree root length density	dimensionless	(1)	Root and Mycorrhiza
303.	Rt_TProxGini	Distribution coefficient of proximal root diameters (CumFreq = (Diam/Diammax) ^{TProxGini} of a tree, used in calculation of the specific root length of a tree root system	dimensionless	0.001 – 10 (.3)	(Tree Library/Roots)
304.	Rt_TWghtDiam1[Tree]	Biomass of a (branched) tree root with a proximal (at stem base) diameter of 1 cm; Intercept (a) of allometric equation (Root weight = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	kg cm ^{b,z}	0.01 – 1 (.5)	(Tree Library/Roots)
305.	Rt_TWghtDiamSlope[Tree]	Power coefficient (b) of allometric equation (RootWeight = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	1 – 3 (2.3)	(Tree Library/Roots)
306.	S&B_2ndFireafterPileup	Number of days between pile up and secondary burn event	days	1 – 100 (5)	Management/Slash and Burn
307.	S&B_CritMoist	Limit value for internal + adhering (intercepted from rainfall) moisture content of slashed necromass; below this value necromass is categorized as dry and fire can take place	l kg ⁻¹	0 – 1(.05)	Management/Slash and Burn

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
308.	S&B_DailyDeadWood-LitTransf	Rate of transfer of daily dead wood litter per day	fraction day ⁻¹	(0.005)	Slash and Burn
309.	S&B_DailyNecromLitTransf	Rate of transfer of necromass litter per day	fraction day ⁻¹	(0.01)	Slash and Burn
310.	S&B_DeathWoodFuelFact	Temperature of the fire per unit dry weight of fuel in dead wood	°C kg ⁻¹	0 – 100 (.1)	Management/Slash and Burn
311.	S&B_FirImpPSorption	Fire impacts on P sorption, as a function of soil surface temperature increase	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
312.	S&B_FirIndPMobiliz	Fire impact on mobilization fraction of P from the inorganic P immobile pool, as a function of soil surface temperature increase	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
313.	S&B_FirMortSeedBank	Fractional mortality in the weed seed bank as a function of soil surface temperature increment	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
314.	S&B_FuelLoadFactor	Temperature of the fire per unit dry weight of fuel in slashed necromass and structural surface litter	°C kg ⁻¹	0 – 100 (10)	Management/Slash and Burn
315.	S&B_MaxDryingPer	The latest time after slashing when fire can occur; if the fuel does not get dry enough before this time, no fire will be occur	days	1 – 200 (30)	Management/Slash and Burn
316.	S&B_MinDryingPer	The earliest time after slashing that fire can occur	days	0 – 100 (20)	Management/Slash and Burn
317.	S&B_NecroBurnFrac	Fraction of surface necromass burnt as a function of fire temperature at the soil surface.	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
318.	S&B_NutVolatFracN	Volatilization fraction of N in the burnt necromass, as a function of soil surface temperature increment	dimensionless	(see table in excel sheet Slash&Burn)	(Slash&Burn)
319.	S&B_NutVolatFracP	Volatilization fraction of P in the burnt necromass, as function of soil surface temperature increment	dimensionless	(see table in excel sheet Slash&Burn)	(Slash & Burn)
320.	S&B_pHRecFrac	Daily recovery fraction of soil pH in the topsoil from its post-fire towards its pre-fire value	fraction	0.001–0.1 (.01)	Management/Slash and Burn
321.	S&B_PileUpFrac	Fraction of dead wood pile up after slash and burn event	fraction	0 – 1 (0.7)	Management/Slash and Burn

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
322.	S&B_PSORPRecFrac	Daily recovery fraction of the P ₂ sorption in the topsoil from its post-fire towards its pre-fire value	fraction	0.001–0.1 (.01)	Management/Slash and Burn
323.	S&B_ScorchWRRemFrac	Fraction of scorched wood removed after slash and burn event	fraction	0 – 1 (.3)	Management/Slash and Burn
324.	S&B_SlashDOY	A graphical input tabulating day of year at which slashing is performed	julian days	(see S&B_Slash-DOY graph)	Management/Slash and Burn
325.	S&B_SlashYear	A graphical input tabulating year at which slashing is performed	dimensionless	any integer value (100)	Management/Slash and Burn
326.	S&B_SOMBurnFrac	Fraction of all SOM pools in the topsoil (Layer 1) respired (C) or mineralized (N & P) as a function of soil surface temperature increment	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
327.	S&B_SurfLitBurnFrac	Fraction of all surface litter respired (C) or mineralized (N & P) as a function of soil surface temperature increment	dimensionless	(see table in excel sheet Slash&Burn)	Management/Slash and Burn
328.	S&B_TimetoPileUp	Number of days between primary burn and pile up (redistribution across the zones) for a secondary burn	days	1 – 100 (15)	Management/Slash and Burn
329.	S&B_TimetoWoodRem	Number of days between primary burn and removal of scorched wood	days	1 – 50 (10)	Management/Slash and Burn
330.	S&B_TTempTol[Tree]	Maximum fire temperature that a tree can tolerate. Temperature above the value will induce tree mortality	°C	40 – 90 (75)	(Tree Library/Slash&Burn)
331.	S&B_WatRetRecFrac	Daily recovery fraction of soil water retention in the topsoil from its post-fire towards its pre-fire value	fraction	0.001 – 0.1 (0.005)	Management/Slash and Burn
332.	S&B_WetnessTempImp	Fractional reduction in fire temperature per unit of moisture content of the fuel	fraction	0 – 1 (.5)	Management/Slash and Burn
333.	S_BDBDRefDecay	Relative rate of decay of the bulk density, returning the surface infiltration rate toward S _{SurfInfiltrPerKsatDef} and the saturated hydraulic conductivity towards S _{KsatDefV}	day ⁻¹	0 – 0.1 (.001)	Soil Structure
334.	S_C_RtStrucFormFrac	Fraction of contribution of crop root decay on root channels	fraction per m	(.1)	Soil Structure
335.	S_KsatDefVi	Saturated hydraulic conductivity of the soil in the absence of macropore structure, as derived from texture-based pedotransfer functions. Read and calculation from Wanukas.xls. Input needed to run pedotranfer refer to the sheet pedotranfer	cm day ⁻¹	1 – 500 (layer 1 = 319, layer 2 = 54, layer 3 = 45, layer 4 = 40)	(Soil Hydraulic)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
336.	S_KsatHperVi	Ratio of saturated hydraulic conductivity in horizontal and vertical direction for layer i	dimensionless	0 – 5 (1)	Soil Structure/K Sat ratio
337.	S_KsatInitVi[Zone]	Saturated hydraulic conductivity of the soil at the macropore structure existing at the start of the simulation. Read and calculation from Wanulcas.xls. Input needed to run pedotransfer refer to the sheet pedotransfer	cm day ⁻¹	1 – 500 (layer 1 = 319, layer 2 = 54, layer 3 = 45, layer 4 = 40)	(Soil Hydraulic)
338.	S_KsatVDeepSub	Saturated hydraulic conductivity of the soil below layer 4, determining the rate of vertical drainage from the soil column	cm day ⁻¹	1 – 100 (20)	Soil Structure
339.	S_RelSurfInfiltrInit[Zone]	Surface infiltration rate at the start of the simulation relative to its default value	dimensionless	100 – 10000 (1000)	Soil Structure
340.	S_RelWormLiti	Relative impact of 'worms' (soil fauna) on increase of saturated hydraulic conductivity in each layer	dimensionless	0 – 1 (1, 0.6, 0.3, 0.1)	Soil Structure
341.	S_RelWormSurf	Relative impact of 'worms' (soil fauna) increase of infiltration rate of the soil surface	dimensionless	0 – 1 (1)	Soil Structure
342.	S_SoilStructDyn?	Switch determining dynamics of soil structure (0 = false, 1 = true) based on decay and re-creation of macropores by soil fauna above the texture-based default values	day ⁻¹	0 or 1 (0)	Soil Structure
343.	S_SurfInfiltrPerKsatDef[Zone]	Ratio of surface infiltration and Ksat for the first soil layer in the default condition of the soil as defined by the pedotransfer function	dimensionless	25 – 10000 (25)	Soil Structure
344.	S_T_RtStrucFormFrac	Fraction of contribution of tree root decay on root channels	fraction per m	(.3)	Soil Structure
345.	S_WormsLikeLitMetab	Activity (in arbitrary units) of soil fauna ("worms") per unit of organic inputs in the litter metabolic pool	m ² kg ⁻¹	0.00001 – 0.1 (.00001)	Soil Structure
346.	S_WormsLikeLitStruc	Activity (in arbitrary units) of soil fauna ("worms") per unit of organic inputs in the litter structural pool	m ² kg ⁻¹	0.0000005 – 0.1 (.0000005)	Soil Structure
347.	S_WormsLikeSOMMetab	Activity (in arbitrary units) of soil fauna ("worms") per unit of organic inputs in the SOM metabolic pool	m ² kg ⁻¹	0.000001 – 0.1 (.000001)	Soil Structure
348.	S_WormsLikeSOMStruc	Activity (in arbitrary units) of soil fauna ("worms") per unit of organic inputs in the SOM structural pool	m ² kg ⁻¹	0.00000005 – 0.1 (.00000005)	Soil Structure
349.	T_ApplyFBA?[Tree]	Switch (1 = yes, 0 = no) to determine whether the allocation of biomass from the canopy to the wood (branches + stem) pools is governed by the fractal branching parameters (allometric equations).	dimensionless	0 or 1 (1)	Tree Parameters

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
350.	T_ApplyPalm?[Tree]	Switch (1 = yes, 0 = no) to determine whether the allocation of biomass to storage pool follows oil palm rule.	dimensionless	0 or 1 (0)	(Tree Library/Fruit)
351.	T_BarkThickness	Bark thickness to calculate target of latex content, the thickness is following the growth of the tree.	cm	0 – 1 (see graph)	Management/Latex Production
352.	T_BrownBast?	Switch (1 = yes, 0 = no) to determine whether the rubber tree infected by brownbast.	dimensionless	0 or 1 (0)	Management/Latex Production
353.	T_CanBiomInit[Tree]	Initial amount of biomass in tree canopy (leaf and small stems)	kg per tree	0 – 1 (0)	Tree parameters
354.	T_CanHMax[Tree]	Maximum height of tree canopy	m	0 – 15 (8.2)	(Tree Library/Canopy)
355.	T_CanMaintResp	Fraction of canopy biomass use for maintenance respiration	dimensionless	(0.001)	Management/Latex Production
356.	T_CanShape[Tree]	Factor determining in which part of the tree leaves are concentrated. A value of 1 gives an even spread of tree leaves over the alley, a higher value (eg 2) concentrates tree leaves above the hedgerow	dimensionless	0 – 2 (.567)	(Tree Library/Canopy)
357.	T_CanWidthMax[Tree]	Maximum tree canopy width, half the canopy width (radius).	m	0 – 10 (4.655)	(Tree Library/Canopy)
358.	T_ConcFruit[SiNut,Tree]	Nutrient concentration in fruit component	dimensionless	0 – 0.1 (N = .02, P = .002)	(Tree Library/N-P concentration)
359.	T_ConcGroRes[SiNut,Tree]	Nutrient concentration in carbohydrate reserves	dimensionless	0 – 0.1 (N = .01, P = .0005)	(Tree Library/N-P concentration)
360.	T_ConcLf[SiNut,Tree]	N concentration in leaf component of tree	dimensionless	0 – 0.1 (N = .0173, P = .0009)	(Tree Library/N-P concentration)
361.	T_ConcRt[SiNut,Tree]	Nutrient concentration in tree roots (only for Rt_ATTtype=2)	dimensionless	0 – 0.1 (N = .0122, P = .0006)	(Tree Library/N-P concentration)
362.	T_ConcTwig[SiNut,Tree]	Nutrient concentration in twig component of tree	dimensionless	0 – 0.1 (N = .00073, P = .0016)	(Tree Library/N-P concentration)
363.	T_ConcWood[SiNut,Tree]	Nutrient concentration in wood component of tree	dimensionless	0 – 0.1 (N = .0047, P = .0008)	(Tree Library/N-P concentration)
364.	T_DiamBiom1[Tree]	Biomass of a tree of diameter 1 cm; Intercept (a) of allometric equation (Branch biomass = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	kg	0.01 – 1 (.6513)	(Tree Library/Allometric branching)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
365.	T_DiamBranch1[Tree]	Intercept (a) of allometric equation (Tree branch biomass = a Diameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	kg	0.01 – 1 (.0334)	(Tree Library/Allometric branching)
366.	T_DiamCumLit1[Tree]	Cumulative litterfall expected for a stem diameter of 1 cm. Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	kg	0.01 – 1 (.0302)	(Tree Library/Allometric branching)
367.	T_DiamLfTwig1[Tree]	Intercept (a) of allometric equation (Leaf & Twig biomass = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	kg cmb	0.01 – 1 (.9656)	(Tree Library/Allometric branching)
368.	T_DiamSlopeBiom[Tree]	Power coefficient (b) of allometric equation (Branch biomass = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	0 – 3 (2.0937)	(Tree Library/Allometric branching)
369.	T_DiamSlopeBranch[Tree]	Power coefficient (b) of allometric equation (Tree branch biomass = a Diameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	0 – 3 (2.4195)	(Tree Library/Allometric branching)
370.	T_DiamSlopeCumLit[Tree]	Power coefficient (b) of the allometric equation describing the increase of cumulative litterfall with stem diameter. Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	0 – 3 (3.0937)	(Tree Library/Allometric branching)
371.	T_DiamSlopeLfTwig[Tree]	Power coefficient (b) of allometric equation (Leaf & Twig biomass = a StemDiameter ^b). Calculation from Functional Branch Analysis (FBA). Input needed to run FBA refer to tree parameterization	dimensionless	1 – 3 (1.7270)	(Tree Library/Allometric branching)
372.	T_DiamTreshHarv[Tree]	Tree diameter of timber harvested	cm	100	Management/Timber Harvesting
373.	T_DOY 1 LfFlush[Tree]	Day of the first cycle of leaf flush	julian days	1 – 365 (1)	Tree parameters/Tree leaf phenology
374.	T_DOY 2 LfFlush[Tree]	Day of the second cycle of leaf flush	julian days	1 – 365 (400)	Tree parameters/Tree leaf phenology
375.	T_DOY SeaLitFall1Start[Tree]	Day when the first season of leaf starts dropdown	julian days	1 – 365 (400)	Tree parameters/Tree leaf phenology
376.	T_DOY SeaLitFall2Start[Tree]	Day when the second season of leaf starts dropdown	julian days	1 – 365 (400)	Tree parameters/Tree leaf phenology

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
377.	T_DOY_Compl1LlFall[Tree]	Day when the first season of leaf completely dropdown	julian days	1 – 365 (400)	Tree parameters/ Tree leaf phenology
378.	T_DOY_Compl2LlFall[Tree]	Day when the second season of leaf completely dropdown	julian days	1 – 365 (400)	Tree parameters/ Tree leaf phenology
379.	T_DOYFlwBeg[Tree]	The earliest day in a year when tree start to flowers	julian days	1 - 365 (200)	(Tree Library/ Growth stage)
380.	T_DOYFlwEnd[Tree]	The latest day in a year when tree start to flowers	julian days	1 - 365 (250)	(Tree Library/ Growth stage)
381.	T_DynTapping Frac	Dynamic value of fraction of latex stock would be tapped everyday. The simulation will run using this dynamic value when T_BrownBast? switch to 1	dimensionless	See graph	Management/Latex Production
382.	T_ExpRetThresh	Threshold value of expected return to labour		(30)	Management/Latex Production
383.	T_FracSealLitFall1[Tree]	Fraction of tree canopy become litterfall	dimensionless	0 – 1 (1)	(Tree parameter/ Tree leaf phenology)
384.	T_FruitAllocMax[Tree]	Allocation of biomass to fruit each day	kg m ² day ⁻¹	0 – 1 (0)	(Tree Library/Fruit)
385.	T_FruitAllocStage[Tree]	Graphical input parameter as a function of tree stage that determine how much fruit will produce from maximum fruit allocation	dimensionless	0 – 1 (see T_FruitAllocStage graph)	Management/Fruit Harvesting
386.	T_FruitHarvFrac[Tree]	Harvest index for fruit. Constant value for every fruiting season	dimensionless	0 – 1 (0)	Management/Fruit Harvesting
387.	T_FruitMoistFrac[Tree]	Standard moisture content for expressing marketable fruit of each tree	dimensionless	0 – 1 (0)	Profitability
388.	T_GenLitFracMax[Tree]	Fraction of fruit will drop	dimensionless	0 – 1 (.05)	Management/Fruit Harvesting
389.	T_GenLitStage[Tree]	Graphical input parameter as a function of tree stage that determine how many fruit will drop	dimensionless	0 – 1 (see T_GenLitStage graph)	Management/Fruit Harvesting
390.	T_GraphPhenol?[Tree]	Parameter governing an option to simulate tree phenology using graph. Value 0 means tree phenology using phenology parameters, value 1 means tree phenology using graph	dimensionless	0 or 1 (0)	Tree parameter/Tree leaf phenology
391.	T_GroMax[Tree]	Maximum growth rate of hedgerows at full canopy closure	kg m ² day ⁻¹	0 – 0.1 (.014)	(Tree Library/ Growth)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
392.	T_GroResFrac[Tree]	Fraction of tree carbohydrate reserves converted to biomass during regrowth stage after pruning	day ⁻¹	0 – 0.5 (.025)	(Tree Library/ Growth)
393.	T_GroResInit[Tree]	Initial amount of tree carbohydrates as reserves of tree potential growth	kg per tree	0 – 1 (0.25)	Tree parameters
394.	T_GrowthResp			(1)	Management/Latex Production
395.	T_HeartWoodAllocAft- Pruned?	Parameter governing an option to simulate heartwood. Value 0 means after pruning no sapwood biomass that allocated to heartwood biomass, value 1 means after pruning all sapwood biomass allocated to heartwood biomass.	dimensionless	(0)	Tree parameter/Tree stem (sapwood & heartwood)
396.	T_InitStage[Tree]	Initial stage of tree when it was planted. If tree already growing at the start of simulation, it is the stage at the start of simulation time	dimensionless	0 – 2 (0)	(Tree Library/Growth stage)
397.	T_KillDOY[Tree]	Schedule date, day of year to kill tree	julian days	1 – 365 (1)	Management/Killing Trees
398.	T_KillY[Tree]	Schedule date, year to kill tree	dimensionless	any integer value (1000)	Management/Killing Trees
399.	T_KLight[Tree]	Tree canopy (leaves Wonen) extinction light coefficient = the efficiency of tree foliage in absorbing light	dimensionless	0 – 1 (.7)	(Tree Parameters/ Light Capture)
400.	T_LAIMax[Tree]	Maximum value of LAI in the tree canopy	dimensionless	0 – 5 (4)	(Tree Library/ Canopy)
401.	T_LAIMinMaxRatio[Tree]	Parameter describing canopy thickness/dense. Value 1 is maximum thickness	dimensionless	0 – 1 (1)	(Tree Library/ Canopy)
402.	T_LatexFormResp	Respiration for Latex formation		(0.01)	Management/Latex Production
403.	T_LatexMainResp	Fraction of latex production use for maintenance respiration	dimensionless	(0.0001)	Management/Latex Production
404.	T_LatexMoistFrac[Tree]	Standard moisture content for expressing marketable latex of each tree	dimensionless	0 – 1 (.14)	Profitability
405.	T_LatexRecoveryTime	Time for latex to recover.		(7)	Management/Latex Production
406.	T_LfallDroughtFrac[Tree]	Fraction of tree biomass becomes litterfall due to drought	day ⁻¹	0 – 1 (.01)	(Tree Library/Lit- terfall)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
407.	T_LifalThreshWStress[Tree]	Threshold value for tree litterfall due to drought	dimensionless	0 – 1 (.7)	(Tree Library/Litterfall)
408.	T_LifalWeight[Zone,Tree]	Input weight value governing amount of tree litterfall going into each zone relative to other zones (eg. 1:1:1:1 means equal mulch given in each zones on area basis)	dimensionless	0 – 10 (1, 1, 1, 1)	Litterfall
409.	T_LignLifal[Tree]	Lignin concentration of tree litterfall (eg. 20%=0.2)	dimensionless	0 – 1 (.43)	(Tree Library/Litter Quality)
410.	T_LignPrun[Tree]	Lignin concentration of pruned tree biomass (eg. 20%=0.2)	dimensionless	0 – 1 (.4)	(Tree Library/Litter Quality)
411.	T_LignRt[Tree]	Lignin concentration of tree root	dimensionless	0 – 1 (.4)	(Tree Library/Litter Quality)
412.	T_LWR[Tree]	Leaf Weight Ratio = leaf dry weight per unit shoot dry weight	dimensionless	0 – 5 (.494)	(Tree Library/Growth)
413.	T_MaxBarkLatexContent	Maximum of latex content in bark	dimensionless	0 – 1 (0.3)	Management/Latex Production
414.	T_MaxGrowthUfFrac			(.2)	Management/Latex Production
415.	T_MaxUseFrac			(.1)	Management/Latex Production
416.	T_MemExpY	How much a farmer forget previous yield (latex yield) and use it as a basis for his expectation of future latex yield, value in 0 - 1. 0 = he remembers fully, 1 = he forgets fully	dimensionless	0 - 1 (0.1)	Management/Latex Production
417.	T_MinDiamforTappingcm[Tree]	Minimum tree diameter for tapping	cm	10 – 15 (15)	Management/Latex Production
418.	T_MycMaxInf[Tree]	Fraction of tree roots infected by mycorrhiza for a soil layer where the Rt_MTInfFrac parameter is 1	dimensionless	0 – 1 (.3)	(Tree Library/Mycorrhiza)
419.	T_NFixDayFrac[Tree]	Fraction of current N deficit derived from atmospheric N ₂ fixation per day for each tree if T_NFixVariable = 1 (('true')	day ⁻¹	0 - 1 (.125)	(Tree Library/N Fixation)
420.	T_NFixDWMMaxFrac[Tree]	Maximum fraction of the T_GroRes[Dw] pool that can be respired for N ₂ fixation if T_NFixVariable = 0 (('false')	day ⁻¹	0 - 0.5 (.05)	(Tree Library/N Fixation)
421.	T_NFixDWWUnitCost[Tree]	Dry weight cost for respiration per unit N ₂ fixation, if T_NFixVariable = 0 (('false')	kg [dw] g ⁻¹ [N]	0 – 1 (0)	(Tree Library/N Fixation)

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No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
422.	T_NfixResp[Tree]	Responsiveness of N ₂ fixation to N stress (N in biomass divided by N target), if T_NFixVariable = 0 ("false")	dimensionless	0 – 5 (0)	(Tree Library/N Fixation)
423.	T_NFixVariable?[Tree]	Switch (0 = false, 1 = true) to choose between variable (N-stress dependent) versus constant N ₂ fixation as fraction of N deficit	dimensionless	0 or 1 (0)	(Tree Library/N Fixation)
424.	T_NlfallRed[SINut,Tree]	Reducing factor for nutrient concentration of tree litterfall which depend on type of tree	dimensionless	0 – 2 (.7 for N and P)	(Tree Library/Litterfall)
425.	T_NutMobT[SINut]	Relative rate of transfer, per unit root length density (cm cm ⁻³), from the 'immobile' pool of nutrients to the 'mobile' or sorbed pool, due to Crop root activity	m ² day ⁻¹	0 – 0.02 (0)	(Tree Library/Root Impacts on Nutrient Mobility)
426.	T_PanelQuality1[Tree]	Bark available for tapping at the first period of tapping			Management/Latex Production
427.	T_PanelQuality2[Tree]	Bark available for tapping at the second period for tapping			Management/Latex Production
428.	T_PanelRecoveryTime	Recovery time of bark have been tapped that would be available for next period of tapping	days	(7300)	Management/Latex Production
429.	T_PlantDOY[Tree]	Schedule for date of planting time. Entered from Wanulcas.xls	julian days	1 – 365 (see table in excel sheet Tree Management)	(Tree Management)
430.	T_PlantY[Tree]	Schedule for year of planting time. Entered from Wanulcas.xls	dimensionless	any integer value (see table in excel sheet Tree Management)	(Tree Management)
431.	T_PolyLifall[Tree]	Polyphenol concentration of tree litterfall (eg. 3 %=0.03)	dimensionless	0 – 1 (0)	(Tree Library/Litter Quality)
432.	T_PolyPrun[Tree]	Polyphenol concentration of pruned tree biomass (eg. 3 %=0.03)	dimensionless	0 – 1 (0)	(Tree Library/Litter Quality)
433.	T_PolyPrt[Tree]	Polyphenol concentration of tree root	dimensionless	0 – 1 (0)	(Tree Library/Litter Quality)
434.	T_PruneDoY[Tree]	Schedule for date of pruning. Entered from Wanulcas.xls	julian days	1 – 365 (365)	(Tree Management)
435.	T_PruneFracC[Tree]	Fraction of canopy that gets pruned, for T_PruneType = 0. Constant for every pruning	dimensionless	0 – 1 (0)	Managements/PruningEvents
436.	T_PruneFracC[Tree]	Fraction of tree canopy gets pruned, for T_PruneFrac? = 0	dimensionless	0 – 1 (1)	Management/PruningEvents

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
437.	T_PrunFracD[Tree]	Fraction of tree canopy that gets pruned, for T_PrunFrac? = 1	dimensionless	0 – 1 (1)	(Tree Management)
438.	T_PrunHarvFracC[Tree]	Fraction of pruned canopy that harvested (not return to the system), for T_PrunType? = 0. Constant for every pruning.	dimensionless	0 – 1 (0)	Managements/PruningEvents
439.	T_PrunHarvFracD[Tree]	Fraction of tree pruned biomass harvested. Value changes overtime	dimensionless	0 or 1 (0)	(Tree Management)
440.	T_PrunLimit	Critical total LAI of all trees shadowing the crop zone, triggering a pruning event	dimensionless	0 – 5 (100)	Management/PruningEvent
441.	T_PrunMoistFrac[Tree]	Standard moisture content for pruned biomass of each tree	dimensionless	0 – 1 (.14)	Profitability
442.	T_PrunPlant?[Tree]	Parameter governing pruning decision. 1 = tree is automatically pruned before crop planting, 0 = tree does not automatically pruned	dimensionless	0 or 1 (1)	Management/PruningEvent
443.	T_PrunRecov[Tree]	Time needed for tree to recover after pruning	days	0 – 30 (14)	Management/PruningEvent
444.	T_PrunStageLimit[Tree]	The latest crop stage at which automatic pruning is still performed. Corresponds to T_PrunPlant? = 1	dimensionless	1 – 2 (1.8)	Management/PruningEvent
445.	T_PrunType?	This parameter govern the type of pruning events. 0 = Pruning determined automatically based on canopy denseness (tree LAI). Associated with T_PrunLimit and Tree_StageLimit. 1 = Pruning determined by calendar. Associated with Pruning section in sheet Tree Management, Wanulcas.xls	dimensionless	0 (0 or 1)	Managements/PruningEvents
446.	T_PrunWeight[Zone,Tree]	Input weight value governing amount of tree pruning going into each zone relative to other zones (eg. equal pruned biomass given in each zones on area basis means 1:1:1:1)	dimensionless	0 – 10 (0, 1, 1, 1)	Management/PruningEvent
447.	T_PrunY[Tree]	Schedule for year of pruning. Entered from Wanulcas.xls	dimensionless	any integer value (100)	(Tree Management)
448.	T_RainWStorCap[Tree]	Rainfall intercepted by tree stored as thin film at leaf surface	dimensionless	(1)	(Tree Library/Rain Interception)
449.	T_RecoveryExp			(.01)	Management/Latex Production
450.	T_RelLatexFormPriority			(.7)	Management/Latex Production
451.	T_RelLightMaxGr[Tree]	Relative light intensity at which shading starts to affect tree growth	dimensionless	0 - 1 (.5)	(Tree Library/Light Capture)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
452.	T_RelPosinZone[Tree]	Position of each tree within each zone; 0 means the position of the trees on the left side of its zone, 1 means on the right side of its zone.	dimensionless	0 – 1 (0)	(AF System)
453.	T_RestDayperTappingDay	Number of days between tapping event	days	(see T_RestDayperTappingDay)	Management/Latex Production
454.	T_RhizEffkP	Proportional reduction of the apparent adsorption constant for P due to root activity of the crop, expressed as fraction of N_Kap-def per unit tree root length density	m ² day ⁻¹	0 – .2 (0)	(Tree Library/Root impacts on P mobility)
455.	T_RtDiam[Tree]	Tree root diameter. It is used in calculating water and nutrient uptake. For all root type.	cm	.05 – 3 (.1)	(Tree Library/Roots)
456.	T_Rubber?	Switch (1 = yes, 0 = no) to determine whether the tree growth as rubber tree, there is allocation of biomass to latex pool	dimensionless	0 or 1 (0)	(Tree Library/Growth)
457.	T_SapWoodScaling Rule	Power coefficient of conversion of diameter of sapwood to heartwood. Default value 1 means no increasing of diameter heartwood	dimensionless	0 – 1 (1)	Tree parameter/Tree stem (sapwood & heartwood)
458.	T_SLA[Tree]	Tree specific leaf area = tree leaf surface area per unit leaf dry weight	m ² kg ⁻¹	0 – 30 (7.87)	(Tree Library/Growth)
459.	T_SlashLabour	Amount of labour involved in slashing the field per unit simulated filed as a function of biomass slashed	person days	(see T_SlashLab graph)	Management/Slash and Burn
460.	T_SlashSellWoodFrac[Tree]	Indicates the fraction of wood that is removed from the plot at the time of slashing the vegetation	dimensionless	0 – 1 (0)	Management/Slash and Burn
461.	T_StageAftPrun[Tree]	Tree growth stage after pruning	dimensionless	0 – 2 (1)	(Tree Library/Growth stage)
462.	T_Tapataill?	Switch (1 = yes, 0 = no) to determine whether the tree will tapped or not	dimensionless	0 or 1 (0)	Management/Latex Production
463.	T_TapGirthFraction	Fraction of girth for tapped	dimensionless	0.3 – 0.5 (0.5)	Management/Latex Production
464.	T_TappableHeight[Tree]	Maximum length of stem bark can be tapped	cm	100 – 150 (100)	Management/Latex Production
465.	T_TappDOW	Tapping day of work	days	380	Management/Latex Production
466.	T_TappingFrac	Constant value of fraction of latex stock would be tapped every day	dimensionless	0 – 1 (0.5)	Management/Latex Production

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
467.	T_TappingFracMultiplier	Quick modification on dynamic value of fraction of latex stock would be tapped	dimensionless	(1)	Management/Latex Production
468.	T_TappingSlice	Bark thickness of tapping	cm	0.3 – 0.5 (0.35)	Management/Latex Production
469.	T_Temp2	Temperature for maintenance respiration	dimensionless	(see T_Temp2 graph)	Management/Latex Production
470.	T_TempRespMaint	A graphical relation between temperature and maintenance respiration	dimensionless	(see T_TempResp- Maint graph)	Management/Latex Production
471.	T_TimeGenCycle[Tree]	Length of generative cycles of tree	days	any integer value (150)	(Tree Library/Growth stage)
472.	T_TimeVeg[Tree]	Length of vegetative cycles of tree	days	any integer value (720)	(Tree Library/Growth stage)
473.	T_TranspRatio[Tree]	Amount of water needed per unit dry matter production of tree	l kg ⁻¹	0 – 500 (300)	(Tree Library/Growth)
474.	T_TranspRatioTime	Graphical input parameter as a function of tree stage that determine the dynamic of tree transpiration per unit biomass production	-	-	Tree Parameter
475.	T_TreesperHa[Tree]	Tree plant density	dimensionless	any integer value (400)	(AF System)
476.	T_WoodBiomInit[Tree]	Initial amount of biomass in tree stem	kg per tree	0 – 1 (0)	Tree Parameters
477.	T_WoodDens[Tree]	Wood density of each tree species	kg m ⁻³	(750)	(Tree Library)
478.	T_WoodFracHRemain	Wood height remain after pruning. If you do not want to harvest wood/timber when pruning makes sure this is a high value, eg. 100.	m	0-100 (100)	Management/Pruning Events/Other Pruning Parameters
479.	T_WoodHarvDOY[Tree]	Schedule for date of pruning. Entered from Wanulcas.xls	julian days	1 – 365 (364)	(Tree Management)
480.	T_WoodHarvFrac[Tree]	Fraction Harvested wood	dimensionless	0 – 1 (.95)	Management/Timber Harvesting
481.	T_WoodHarv[Tree]	Schedule for year of timber harvesting. Entered from Wanulcas.xls	dimensionless	Any integer value (100)	(Tree Management)
482.	T_WoodHInit[Tree]	Initial value of tree bare stem height (tree height excluded canopy)	m	0 – 15 (0)	Tree Parameters

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
483.	T_WoodMaintResp	Fraction of wood biomass use for maintenance respiration	dimensionless	(0.000001)	Management/Latex Production
484.	T_WoodMoistFrac[Tree]	Standard moisture content for expressing marketable wood of each tree	dimensionless	0 – 1 (0)	Profitability
485.	Temp_AType	A number governing type of soil temperature data used in the simulation(0= constant value of soil temperature , 1 =read from monthly average data,2=read from daily data which is read from external file)	dimensionless	1, 2 or 3 (1)	Soil Temperature
486.	Temp_Const	Soil temperature throughout the simulation; corresponds to Temp_AType=0	0 ^o	15 – 40 (28)	Soil Temperature
487.	Temp_DailyData	Actual daily data of soil temperature; corresponds to Temp_AType=2. Read from Wanulcas.xls	0 ^o	(see table in excel sheet weather)	(WEATHER)
488.	Temp_DailyPotEvap	Daily potential evaporation. Entered from Wanulcas.xls	mm day ⁻¹	(see table in excel sheet weather)	(WEATHER)
489.	Temp_EvapPotConst	Amount of water evaporating from top soil in absence of plant cover	mm day ⁻¹	0 – 10 (3)	Soil Evaporation
490.	Temp_MonthAvg	Monthly average of soil temperature; corresponds to Temp_AType=1. Entered as graphical function	0 ^o	(see Temp_MonthAvg graph)	Soil Temperature
491.	Temp_PotEvapConst?	Parameter governing type of soil evaporation potential data. 1 = constant throughout simulation, 0 = daily data, 2 = monthly data based on thornthwaite calculation	dimensionless	0 or 1 (1)	Soil Evaporation
492.	TW_DrivenEnergyEpot?	Switch (1 = yes, 0 = no) to determine whether the tree water demand driven by Epot	dimensionless	(0)	Soil Evaporation
493.	TW_PotSuctAlphaMax[Tree]	Plant potential where transpiration is (1-Alpha)*potential transpiration, where Alpha is a small value (e.g. 0.01)	cm	-7000 – -3 000 (-5000)	(Tree Library/Water Uptake)
494.	TW_PotSuctAlphaMin[Tree]	Plant potential where transpiration is Alpha*potential transpiration, where Alpha is a small value (e.g. 0.01)	cm	-30000 – -10000 (-15000)	(Tree Library/Water Uptake)
495.	W_FieldCapKcriti[Zone]	Field capacity determined by a threshold rate of subsequent drainage (Kcrit) that is set in the pedotransfer worksheet; the actual field capacity used is the maximum of this value and the field capacity derived from the height above a groundwater table. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	m ³ water m ⁻³ soil	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)

No.	Acronym	Definition	Dimensions	Range of value (Default value)	Input Section (Link location in Excel)
496.	W_Hyd?	Parameter governing water hydraulic lift application in model. 1= apply hydraulic lift in overall water balance, 0=otherwise	dimensionless	0 – 1 (0)	Run & Output Section
497.	W_HydEqFraction	Fraction of water hydraulic lift	fraction	(.1)	Soil Water and Nutrient
498.	W_PhiPxi[Zone]	Graphs showing relationship between pressure head in i-th soil layer of each zone and matrix flux potential (the index x refers to the plants with the highest (H), lowest (L), medium-high (MH) or medium low (ML) rank of root water potential), but the graphs will be identical. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	cm ² day ⁻¹	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)
499.	W_PhiTheta[Zone]	Matrix flux potential at a given theta/soil water content in layer i of each zone. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	cm ² day ⁻¹	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)
500.	W_PTtheta[Zone]	Graphs showing relationship between volumetric soil water content and pressure head in i-th soil layer of each zone. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	cm	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)
501.	W_ThetaInacci[Zone]	Amount of volumetric soil water in i-th soil layer of each zone not available for plant. It is value of volumetric soil water at pF= 4.2 or P = -16000. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	l m ² day ⁻¹	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)
502.	W_ThetaIniti[Zone]	Initial volumetric soil water content related to water saturated pore volume in i-th soil layer of each zone. Current values are 1, 0.9, 0.8, 0.7 for layer 1...4, respectively	fraction	0 – 1	Soil Water and N/ Initial Soil Water
503.	W_ThetaPi[Zone]	Graphs showing relationship between pressure head in i-th soil layer of each zone and volumetric soil water content. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	cm	0 – 0.5 (see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)
504.	W_ThetaPMax[Zone]	Volumetric soil water content at a given maximum soil potential at top layer. Read and calculation from Wanulcas.xls (pedotransfer). Input needed to run pedotransfer refer to the sheet pedotransfer	cm	(see table in excel sheet Soil Hydraulic)	(Soil Hydraulic)

Appendix 8. Statistical criteria for model evaluation result according to Loague and Green (1991)

Criterion	Symbol	Calculation formula	Range	Optimum
Maximum error	ME	$Max \left P_i - O_i \right _{i=1}^n$	≥ 0	0
Root mean square	RMSE	$\left(\sum_{i=1}^n \frac{(P_i - O_i)^2}{n} \right)^{\frac{1}{2}} * \frac{100}{O_{mean}}$	≥ 0	0
Coefficient of determination	CD	$\frac{\sum_{i=1}^n (O_i - O_{mean})^2}{\sum_{i=1}^n (P_i - O_{mean})^2}$	≥ 0	1
Modelling efficiency	EF	$\frac{\left(\sum_{i=1}^n (O_i - O_{mean})^2 - \sum_{i=1}^n (P_i - O_i)^2 \right)}{\sum_{i=1}^n (O_i - O_{mean})^2}$	≤ 1	1
Coefficient of residual mass	CRM	$\frac{\left(\sum_{i=1}^n O_i - \sum_{i=1}^n P_i \right)}{\sum_{i=1}^n O_i}$	≤ 1	0

P_i = predicted values, O_i = observed values, n = number of samples and O_{mean} is the mean of the observed data.

Appendix 9. Other Useful parameters and their definition (parameters which can be input or output which are not yet at users interface layer)

No	Acronym	Definition	Dimensions	Default value	Model Sector
1.	C_GroResMobFrac	Fraction of crop growth reserves that is remobilize into yield and root component	dimensionless	0.95	Crop Growth
2.	C_NDemand	Amount of Nitrogen or Phosphorus demanded by crop per day; for a crop biomass less than 2 Mg/ha demand is based on 5% N _i above that (after canopy closure) on 1% N	kg ha ⁻¹	output	Crop Growth
3.	C_NUptTot	Total amount of Nitrogen or Phosphorus in all soil layers potentially available for crop uptake per day	g m ⁻²	output	Crop Growth
4.	CW_DemandAct	Actual amount of crop water demand per day. Potential demand reduced by plant water potential.	l m ⁻²	output	Crop Water
5.	CW_UptTot	Total amount of water uptake by crop from all soil layers per day in each zone	l m ⁻²	output	Crop Water
6.	C_SeedConc[PlantComp]	Nutrient concentration on seed. Value for DW =1, N = 0.05 and P = 0.005	-	see definition	Crop Growth
7.	Evap_Surf	Amount of water evaporates from top soil per day in each zone	l m ⁻²	output	Soil Evaporation
8.	GHG_GWP_CH4	Global Warming Potential for CH ₄ . Global Warming potential is the warming effect of a trace gas on the atmosphere. It differs between trace gases due to different atmospheric lifetime and different heat absorption capacity. It is expressed per molecule basis relative to CO ₂	-	15	GHG & Denitrification
9.	GHG_GWP_NO2	Global Warming Potential for NO ₂	-	310	GHG & Denitrification
10.	Mc_LitterC	Amount of carbon in dead plant biomass	g m ⁻²	output	Litter C
11.	Mn_LitterNlnpN [Zone, SiNut]	Amount of Nitrogen or Phosphorus in litter layer	g m ⁻²	output	Litter N
12.	N_ExcLittLay1 [Zone, SiNut]	Mineralisation Nitrogen or Phosphorus in litter layer	g m ⁻²	output	N Layer 1
13.	N_SomMini Exch [Zone, SiNut]	Mineralisation of Nitrogen or Phosphorus in soil layer i	g m ⁻²	output	N Layer 2-4
14.	Rt_CLrvt	Crop root growth as a function of time	cm cm ³ day ⁻¹	output	Crop Root
15.	S_BDEqPower	Power coefficient (b) of the allometric equation describing the decreasing of bulk density later effect on surface infiltration and saturated hydraulic conductivity	dimensionless	0.5	Soil Structure Dynamic
16.	T_BiomAllTrees	Total amount of aboveground biomass for all trees	kg m ⁻²	output	Tree Growth

No	Acronym	Definition	Dimensions	Default value	Model Sector
17.	T_HarvPrunCum[Tree]	Total pruned tree biomass harvested	kg m ⁻²	output	Tree Growth
18.	T_LifallCum[Tree]	Cumulative amount of tree litterfall	kg m ⁻²	output	Tree Growth
19.	T_PrunCum	Cumulative amount of tree pruned biomass	kg m ⁻²	output	Tree Growth
20.	T_WatStressMem	A parameter influencing the effect of drought on litterfall.	-	0.75	Tree Water Parameter
21.	TF_AbReSizePow[Tree]	Input parameter in oil palm module	-	2	Tree Fruit
22.	TF_AvgDWpFruit[Tree]	Average dry weight of oilpalm fruit	kg m ⁻²	-	Tree Fruit
23.	TF_FemSinkperBunch	Graphical input parameter in oil palm module as a function of fruit stage	-	-	Tree Fruit
24.	TF_FirstBudtoFlowerInit [Tree]	Number of phyllochron time units before sex determination during flower development	-	25	Tree Fruit
25.	TF_InitFruitpBranch [Tree]	Input parameter in OilPalm module	-	500	Tree Fruit
26.	TF_MaleSinkperBunch	Graphical input parameter in oil palm module as a function of fruit stage	-	-	Tree Fruit
27.	TF_MaleThresh[Tree]	Input parameter in OilPalm module	-	0.17	Tree Fruit
28.	TF_MTresptowStress[Tree]	Input parameter in OilPalm module	-	0	Tree Fruit
29.	TF_PhyllocronTime[Tree]	Input parameter in oil palm module	-	15	Tree Fruit
30.	TF_PhysAgeInit[Tree]	Phyllochron age at time of planting or start of simulation. Input parameter in oil palm module	-	15	Tree Fruit
31.	TF_StageAbortSens	Graphical input parameter in oil palm module as a function of fruit stage	-	-	Tree Fruit
32.	TF_WatStressAbortFrac[Tree]	Input parameter in OilPalm module	-	0.1	Tree Fruit
33.	TW_Alpha	A small value determining plantt maximum and minimum transpiration. Plant potential for maximum transpiration'(TW_PotSuctAlphaMax, unit in cm) is defined as (1- TW_Alpha)*potential transpiration and plant potential for minimum transpiration'(TW_PotSuctAlphaMin, unit in cm) is defined as TW_Alpha*potential transpiration.	-	0.1	Tree Water
34.	TW_AxResFactor	For a 10 m distance and a demand of 5 mm day ⁻¹ we expect the need for an additional gradient of 2.5 bar = 2500 cm; the factor is thus 5/250 = 0.02	-	0.02	Tree Water
35.	TW_L	Coefficient related to tree root conductivity	cm day ⁻¹	10 ⁻⁵	Tree Water
36.	TW_PotSuctBuff	Hydrostatic gradient to overcome transport resistance in wet soil l	cm	0.05	Tree water parameter
37.	W_PMax	Maximum value of soil potential in positive value	cm	0	Water layer 1
38.	W_SeepScalar	A constant that determine how much water will infiltrate to the deeper soil layer when soil water content go beyond field capacity.	fraction	0	Water layer 1...4